Subject: PHYSICS (042)
Sample Question Paper
Class – XII
Academic Session 2026–27

Maximum Marks: 70 **Time Allowed: 3 hours**

General Instructions

(1) There are 33 questions in all. All questions are compulsory.

(2) This question paper has five sections: Section A, Section B, Section C, Section D and Section E.

(3) All the sections are compulsory.

(4) Section A contains sixteen questions, twelve MCQ and four Assertion-Reasoning based of 1mark each, Section B contains five questions of two marks each, Section C contains seven questions of three marks each, Section D contains two case study-based questions of four marks each and Section E contains three long answer questions of five marks each.

(5) There is no overall choice. However, an internal choice has been provided in two questions in Section B, one question in Section C, and all three questions in Section E. You have to attempt only one of the choices in such questions.

(6) Use of calculators is not allowed.

(7) You may use the following values of physical constants where ever necessary.

i. $c = 3 \times 10^8$ m/s

ii. $m_e = 9.1 \times 10^{-31}$ kg

iii. $m_p = 1.7 \times 10^{-27}$ kg

iv. $e = 1.6 \times 10^{-19}$ C

v. $\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1}

vi. $h = 6.63 \times 10^{-34}$ J s

vii. $\varepsilon_0 = 8.854 \times 10^{-12} C^2 N^{-1} m^{-2}$

viii. Avogadro's number = 6.023×10^{23} per gram mole

ix. radius of nucleus $\approx 10^{-15}$ m

	SECTION A	**(16 x 1 = 16)**
Q.No.	**Question**	**Marks**
1.	A point charge **+Q** is placed at the centre of a spherical Gaussian surface of radius **R**. If the radius of the sphere is doubled, the electric flux through the surface will A. become half B. become double C. remain the same D. become four times	1

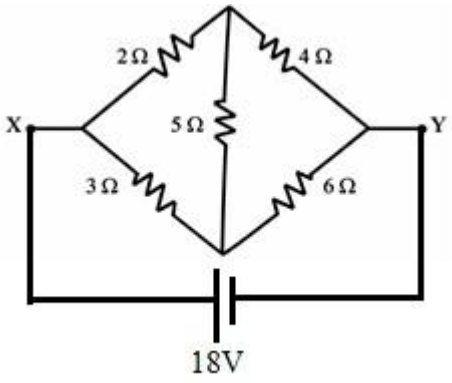
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2.	If two identical heaters each rated as (1000 W, 220 V) are connected in parallel to 220 V, then the total power consumed is- A. 200 W B. 250 W C. 2000 W D. 2500 W	1
3.	An infinitely long cylinder is kept parallel to a uniform magnetic field B directed along negative y –axis. What will be the direction of induced current as seen from the y axis? A. clockwise of the negative y-axis B. anticlockwise of the negative y-axis C. no current will be induced D. along the direction of the magnetic field	1
4.	If electric field associated with an electromagnetic wave is given by **E**= 30 sin $(0.2\pi \times 10^{-4}\ m^{-1}x - 0.6\ \pi \times 10^{4}\ Hz\ t)\hat{j}$ then the direction of propagation of the wave is along, A. $\hat{\imath}$ B. $-\hat{\imath}$ C. $\hat{k}$ D. $-\hat{k}$	1
5.	Which one of the following phenomenon is **not** explained by Huygens' construction of wave front? A. Diffraction B. Refraction C. Reflection D. Origin of spectra	1
6.	If wave front from star in nearby galaxy is detected by a telescope at the top of some mountain in India then the shape of the wavefront detected will be A. diverging spherical B. converging spherical C. plane D. cylindrical	1

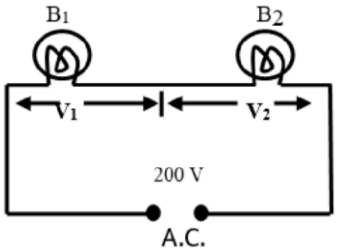
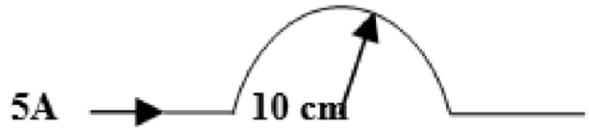
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7.	The binding energy per nucleon curve shows a maximum around mass number A=56. What does this imply about nuclei near iron (Fe)? A. They are the least stable nuclei. B. They are the most tightly bound nuclei. C. They undergo spontaneous fission releasing maximum energy. D. They cannot participate in nuclear reactions.	1
8.	A ball of superconducting material is dipped in liquid nitrogen and placed near a bar magnet. The direction of the movement of the ball and the magnetic moment would be-	1

	Direction of the movement of ball	Direction of the magnetic moment
A.	Towards the bar magnet	Opposite to that of the bar magnet
B.	Towards the bar magnet	Same direction as that of the bar magnet
C.	Away from the bar magnet	Same direction as that of the bar magnet
D.	Away from the bar magnet	Opposite to that of the bar magnet

9.	A bridge circuit is shown in the following figure. The total current flowing in the circuit is-  A. 18 A B. 5 A C. 5/18 A D. 18/5A **For Visually Impaired Students only** In a Wheat-stone bridge, all the four arms have equal resistance R. If the resistance of the galvanometer arm is 2R, the equivalent resistance of the combination as seen by the battery is- A. 2R B. R C. R/2 D. R/4	1

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10.	Two electric bulbs, B_1 and B_2, are connected in series to an A.C. supply of 200 V. The power ratings printed on the bulbs are:  • B_1: 200 V, 60 W • B_2: 200 V, 100 W Using their rated voltage–power information, the ratio of their resistances R_1/R_2 is- A. 5:3 B. 3:5 C. 3:4 D. 4:3	1
11.	If nucleus scaled to football size is approximately 22cm, then the radius of the orbit of electron will be approximately: A. 22 cm B. 22 m C. 22 km D. 22000 km	1
12.	A current of 5A is passing through a long wire which has semicircular loop of radius 10 cm as shown in the figure-  The magnetic field produced at the center of the loop is- A. 2 π μ T B. 4 π μ T C. 5 π μ T D. 8 π μ T	1

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	For Questions 13 to 16, two statements are given –one labelled Assertion (A) and other labelled Reason (R). Select the correct answer to these questions from the options as given below.	
	A. both Assertion and Reason are true and Reason is the correct explanation of Assertion. B. both Assertion and Reason are true but Reason is not the correct explanation of Assertion. C. Assertion is true but Reason is false. D. both Assertion and Reason are false.	
13.	**Assertion** (A): In Rutherford's scattering experiment, most of the alpha particles passed through the gold foil undeflected. **Reason** (R): The electrons in the atom are very light and do not affect the path of the alpha particles significantly.	1
14.	A convex lens of focal length 20 cm is used to form a real image of a point object placed 30 cm from the lens. The lens is then immersed in water of refractive index 1.33 (the lens material has refractive index 1.5). **Assertion** (A): The focal length of the lens increases when it is immersed in water. **Reason** (R): The focal length of a lens depends on the relative refractive index of the lens material with respect to the surrounding medium, as given by the lens maker's formula.	1
15.	In Young's double slit experiment, the distance between the slits is 0.25mm and the screen is placed 1.0m away. A student uses monochromatic light of wavelength 500nm. **Assertion** (A): The fringe width observed on the screen will be 2.0mm. **Reason** (R): The fringe width in YDSE is directly proportional to the slit separation and inversely proportional to the wavelength of light.	1
16.	**Assertion** (A): If a conductor is placed in an external electric field, the electric field inside the conductor becomes infinity in electrostatic equilibrium. **Reason** (R): Free electrons inside the conductor rearrange themselves in such a way that the internal electric field is in the same direction as that of the external field.	1

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	SECTION B	
17.	The electric and magnetic fields associated with an electromagnetic radiation in a medium are given as follows: $E = 30 \sin (2\pi \times 10^{18} \text{rads}^{-1} t + \pi \times 10^{10} \text{m}^{-1} x)$ Vm^{-1} $B = 10^{-7} \sin (2\pi \times 10^{18} \text{rad}^{-1} t + \pi \times 10^{10} \text{m}^{-1} x)$ T I. What is the refractive index of the medium through which the wave is propagating? II. Find the frequency of the wave and identify the electromagnetic wave.	2
18.	Write the nature of path of free electrons in a conductor in the a) presence of electric field, b) absence of electric field.	2
19.	Two like charges +q and +q are placed at a distance d. What work must be done by an external agent to bring another charge +q from infinity to the midpoint of the line joining the two given charges?	2
20(A)	A certain magnetic substance is found to have a relative permeability of 800. What type of magnetic material is it? Write any two characteristics that such a material exhibits. **OR**	2
20(B)	State Gauss's law of magnetism. Also discuss its significance.	
21(A)	I. Why hydrogen spectral lines are considered to be a fingerprint of the element? II. If energy of an electron of the hydrogen atom in ground state is -13.6 eV, what should be the minimum energy of photon which can ionize the atom? **OR**	2
21(B)	State Bohr's quantization condition for defining stationary orbit. How does de Broglie hypothesis explain the stationary orbit?	
	SECTION C	
22.	Compare the energy band diagrams of metals, insulators, and semiconductors. I. Explain how the band gap determines conductivity. II. Apply this reasoning to justify why diamond (C) is an insulator while tin (Sn) is a conductor, though both are the elements of group IV.	3

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23.	Explain- I. It is easier to start a car engine on a warm day than on a chilly day. II. The resistance of our body is so large, even then one experiences a strong shock (sometimes even fatal)) when one accidentally touches the live wire of, say 240 V supply. III. Heat is generated continuously in an electric heater but its temperature becomes constant after some time.	3
24.	The atomic mass of $^{16}_{8}O$ is 15.99493 u. Calculate the binding energy per nucleon of the oxygen nucleus. *(Given: mass of proton = 1.00727 u, mass of neutron = 1.00866 u, 1 u = 931.5 MeV/c^2)*	3
25.	A diamond has a refractive index of 2.42. I. A ray of light inside the diamond strikes the surface at an angle of incidence of 30^0 .State whether it will undergo refraction or total internal reflection. II. Explain why diamonds sparkle when cut properly.	2+1
26.	Derive Snell's law of refraction using Huygens' wave theory. What prediction of the wave theory was later experimentally verified?	3
27(A)	I. An ammeter has an internal resistance of 13 Ω and is originally designed to measure currents only up to 100 A**.** To extend its range for high-current measurements, a shunt resistor is connected in parallel with it. After adding the shunt, the same meter can now read currents as high as 750 A without damage. Using the principle of current division in parallel circuits, determine the value of the shunt resistance required to increase the measuring range of the ammeter. II. The coil of a moving coil galvanometer is wound over a metal frame. Why?	3
	OR	
27(B)	I. A small current loop behaves like a tiny magnet when placed in a magnetic field. Using this idea, explain what physical quantity determines the strength and orientation of such a magnetic dipole. Also state the SI unit of this quantity. II. A steel wire of length l has a magnetic moment m. It is bent into a semicircular arc. What is its new magnetic moment?	

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28.	I. A bar magnet falls from a height through a metal ring. Its acceleration will be less than 'g'. Explain. II. A coil of copper wire is wound uniformly on a wooden equilateral triangular frame. If each side of the triangle is increased to 5 times its original length, while keeping the number of turns per unit length unchanged, how does the self-inductance of the coil change?	3
	SECTION D	
29.	In 2023, researchers developed flexible semiconductor films using organic polymers that could be printed like newspaper sheets. These films can bend, fold, and still conduct electricity efficiently. Unlike traditional silicon chips, polymer semiconductors are lightweight, biodegradable, and can be integrated into clothing or medical patches. Imagine a shirt that monitors your heartbeat or a bandage that tracks wound healing in real time. Such innovations extend the idea of semiconductors beyond rigid circuits, showing how manipulating band gaps and carrier mobility in new materials can revolutionize electronics for healthcare, sustainability, and wearable technology.	
	I. Which advantage of semiconductors makes them ideal for wearable devices like smart shirts? A. High voltage operation (~100 V) B. Bulky design and short life C. Low power consumption and reliability D. Requirement of external heating	1
	II. Carbon (diamond) has a band gap of 5.4 eV, while silicon has 1.1 eV. Why is silicon used in electronic devices but not the diamond? A. Diamond has too many free electrons. B. Diamond's band gap is too large for thermal excitation at room temperature. C. Silicon has weaker covalent bonds than diamond. D. Diamond cannot form covalent bonds.	1
	III. In an intrinsic semiconductor at room temperature A. only electrons contribute to conduction. B. only holes contribute to conduction. C. number of electrons equals number of holes. D. no conduction occurs at all.	1
	IV. How will you use the concept of majority and minority carriers to explain the efficiency of polymer semiconductors in wearable technology?	1

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30.	In 1905, Albert Einstein explained the photoelectric effect by proposing that light consists of discrete packets of energy called photons. Each photon carries an energy $E=h\nu$. When a photon strikes a metal surface, it can transfer its energy to an electron. If the photon's energy exceeds the work function of the metal, the electron can be emitted with maximum kinetic energy $K_{max}=h\nu-\phi_0$. This discovery challenged the classical wave theory, which predicted that electron emission should depend only on the intensity of light. Instead, experiments showed that emission of electrons depends on frequency too, proving the particle nature of light.	
	I. Define the term *work function*.	1
	II. Why the increase in the intensity of light does not increase the maximum kinetic energy of photoelectrons?	1
	III. A metal surface has a work function of 2 eV. The light of frequency 1.2×10^{15}Hz falls on it. Which statement is correct? A. No electron is emitted because the intensity is low. B. Electrons are emitted with maximum kinetic energy less than 2 eV. C. Electrons are emitted with maximum kinetic energy greater than 0 eV. D. Electron emission depends only on the intensity of light and not on the frequency.	1
	IV. Which observation/s directly contradicts the classical wave theory of light? A. Photoelectric current increases with intensity B. The maximum kinetic energy depends only on the frequency and not on the intensity. C. The electrons are emitted instantaneously. D. Both B and C.	1
	SECTION E	
31(A)	I. An isolated conductor cannot have a large capacitance. Why? II. How would you connect 6 μF, 9 μF and 18 μF capacitors to obtain- i. the minimum capacitance ii. the maximum capacitance III. The capacitance of a parallel plate capacitor is increased as the area of the plates increases. What are the other two factors on which the capacitance depends?	2+2+1
	OR	

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31(B)	I. Show that the electric field at any point is equal to the negative of the potential gradient at that point. II. A small metallic sphere is gradually charged in open air. It is observed that electrical breakdown of air occurs when the electric field at its surface reaches 2.5×10^6 V m^{-1} .If the sphere is allowed to attain a maximum potential of 5×10^5 V without causing any discharge, determine the minimum radius the insulated sphere must have.	3+2
32(A)	A compound microscope consists of an objective lens of focal length 2 cm and an eyepiece lens of focal length 5 cm. The object is placed 2.2 cm from the objective lens. I. Calculate the position of the image formed by the objective lens. II. The eyepiece is so placed that the final image is formed at infinity. Find the magnifying power of the microscope. III. Explain briefly why the compound microscope provides much higher magnification compared to a simple magnifier.	2+2+1
	OR	
32(B)	An astronomical telescope uses an objective lens of focal length 100 cm and an eyepiece of focal length 5 cm. I. Calculate the magnifying power of the telescope when the final image is formed at infinity. II. If the least distance of distinct vision of the observer is 25 cm, calculate the magnifying power when the final image is formed at this distance. III. Explain why the objective lens of an astronomical telescope is made with a large focal length and aperture, while the eyepiece has a small focal length.	
33(A)	A certain electrical device is used to convert an alternating voltage of smaller magnitude into a much larger alternating voltage without violating the law of conservation of energy. I. With the help of a neat labelled diagram, describe the principle on which this device operates and explain how it increases the voltage. II. State the basic working mechanism of the device and mention the condition required in its input signal for proper functioning. III. Give one reason why this device does not operate with 100% efficiency in practical situations.	2+2+1
	OR	
33(B)	I. What do you mean by the resonance condition of a series LCR–circuit? II. Write an expression for the resonant frequency. Describe the use of a series resonant circuit in the tuning of a radio receiver.	3+2

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CHEMISTRY (CODE - 043)

SAMPLE QUESTION PAPER*

CLASS XII (2026-27)

Time: 3 hours **Max. Marks: 70**

GENERAL INSTRUCTIONS:

Read the following instructions carefully.

1. There are **33** questions in this question paper with internal choice.
2. SECTION A consists of 16 multiple-choice questions carrying 1 mark each.
3. SECTION B consists of 5 very short answer questions carrying 2 marks each.
4. SECTION C consists of 7 short answer questions carrying 3 marks each.
5. SECTION D consists of 2 case-based questions carrying 4 marks each.
6. SECTION E consists of 3 long answer questions carrying 5 marks each.
7. All questions are compulsory.
8. In addition to this, a separate question has been provided for visually impaired candidates in lieu of questions having visual inputs.
9. Use of log tables and calculators is not allowed.

Section-A

Question 1 to 16 are multiple choice questions. Only one of the choices is correct. Select and write the correct choice as well as the answer to these questions.

1 The boiling points and K_b values of isomeric amines are given in table 1. **1**

Amine	Boiling point/K	pK_b (approx)
P	329.3	3.00
Q	310.5	9.83
R	350.8	3.22

Table 1

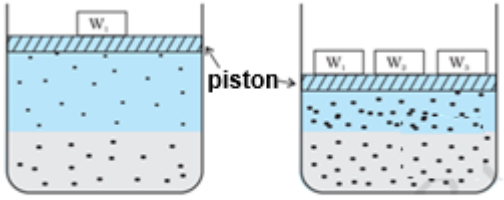
Which one of the following statement is correct?

A. Amine P is more volatile than amine R, but less basic
B. Amine R will give white precipitate with Hinsberg reagent, which is soluble in NaOH
C. Amine Q is more soluble in water as compared to R.
D. Amine P and Q can be prepared by Gabriel Phthalimide process.

2 Which of the following complex is tetrahedral in shape? **1**

A. $[Ni(CN)_4]^{2-}$
B. $[Cu(NH_3)_4]^{2+}$
C. $[NiCl_4]^{2-}$
D. $[Co(en)_3]^{3+}$

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3	The quantity of charge required to obtain one mole of metal "M" from M_2O_3 is: A. 1F B. 6F C. 3F D. 2F	1
4	An organic compound "X" on reaction with chloroform and alcoholic KOH gives foul smelling product. In which of the following case compound X would be produced: A. Chloromethane is treated with AgCN. B. Nitrobenzene is reduced. C. Benzene diazonium chloride is treated with Cu/KCN D. Aniline reacts with Acetyl chloride.	1
5	Consider a system in a state of dynamic equilibrium, as shown in Fig. 1(a). The lower part is solution and the upper part is gaseous system at pressure p and temperature T. If the pressure is increased over the solution phase by compressing the gas to a smaller volume as shown in Fig 1(b).  Figure 1 Which of the following statements is true about figure 1(b) A. solubility of gas will increase until dynamic equilibrium is reached B. solubility of gas will decrease until dynamic equilibrium is reached C. solubility of gas will increase until static equilibrium is reached D. solubility of gas will decrease until static equilibrium is reached ***(For Visually Challenged Learners)*** According to Henry's law partial pressure P of a gas above the liquid at constant temperature is directly proportional to A. volume of the gas B. mole fraction of the gas in the solution C. atmospheric pressure D. vapour pressure of the solution	1
6	Aavya carries out hydrolysis of acetyl chloride. The reaction is represented by the following equation: $CH_3COCl + H_2O \longrightarrow CH_3COOH + HCl$ She carries out the reaction with (i) equimolar amounts of acetyl chloride and water (ii) excess of water. The rate law obeyed in the two cases will be: A. (i) First order (ii) Second Order B. (i) First order (ii) First order C. (i) Second order (ii) Second order D. (i) Second order (ii) First order	1

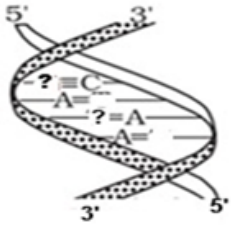
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7	The amino acid HOOC-CH_2-CH_2-CH (NH_2) COOH, can be synthesised within our body. Which of the following is true about this amino acid? HOOC-CH_2-CH_2-CH(NH_2) COOH is: A. an acidic and non-essential amino acid. B. a neutral and non- essential amino acid. C. an acidic and essential amino acid D. a neutral and essential amino acid	1
8	Identify the Grignard reagent and carbonyl compound which on reaction followed by hydrolysis will give 3-methyl pentan-3-ol. (i) $C_2H_5COC_2H_5 + CH_3MgCl$ (ii) $C_2H_5COCH_3 + C_2H_5MgCl$ (iii) $CH_3CHO + C_2H_5 - \underset{\displaystyle MgCl}{\overset{\displaystyle H}{\overset{|}{C}}} - C_2H_5$ (iv) $C_2H_5CHO + CH_3 - \underset{\displaystyle MgCl}{\overset{\displaystyle H}{\overset{|}{C}}} - C_2H_5$ A. (i) and (iii) B. (ii) and (iv) C. (iii) and (iv) D. (i) and (ii)	1
9	Which one of the following element will be strong oxidising agent in their +3 oxidation state? (Atomic no. Cr=24, Mn=25, Ti=22, Fe=26) A. Cr B. Mn C. Ti D. Fe	1
10	When (–)-2-bromohexane reacts with sodium hydroxide, (+)-hexan-2-ol is formed. Which one of the following observation would be correct? A. The reaction follows S_N2 mechanism and is accompanied by inversion of configuration. B. The reaction follows S_N1 mechanism and is accompanied by inversion of configuration. C. The reaction follows S_N2 mechanism and is accompanied by retention of configuration. D. The reaction follows S_N1 mechanism and is accompanied by retention of configuration.	1
11	If half-life of a first order reaction is 16 minutes, how much time does it take to complete 75% of the reaction? A. 16 min B. 32 min C. 48 min D. 64 min	1
12	Which solution has the highest conductivity? A. 0.1 M HCl B. 0.1 M CH_3COOH C. 0.1 M glucose D. 0.1 M NH_4OH	1

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13	**Assertion (A):** Nitration of aniline gives significant amount of meta derivative. **Reason (R):** -NH_2 is ortho and para directing and highly activating group. Select the most appropriate answer from the options given below: A. Both A and R are true, and R is the correct explanation of A. B. Both A and R are true, and R is not the correct explanation of A. C. A is true but R is false. D. A is false but R is true.	1
14	**Assertion (A):** Permanganate and manganate ions have tetrahedral structure **Reason (R):** In permanganate and manganate ions, pi bonding takes place which is due to overlap of p orbitals of oxygen with p orbitals of manganese. Select the most appropriate answer from the options given below: A. Both A and R are true, and R is the correct explanation of A. B. Both A and R are true, and R is not the correct explanation of A. C. A is true but R is false. D. A is false but R is true.	1
15	**Assertion (A)**: Miscibility of ethers with water resembles those of alcohols of the same molecular mass. **Reason (R)**: The oxygen of ether can also form hydrogen bonds with water. Select the most appropriate answer from the options given below: A. Both A and R are true, and R is the correct explanation of A. B. Both A and R are true, and R is not the correct explanation of A. C. A is true but R is false. D. A is false but R is true	1
16	**Assertion (A):** $[Co(NH_3)_6]^{3+}$ is diamagnetic. **Reason (R):** NH_3 is a strong field ligand and causes pairing of electrons. Select the most appropriate answer from the options given below: A. Both A and R are true, and R is the correct explanation of A. B. Both A and R are true, and R is not the correct explanation of A. C. A is true but R is false. D. A is false but R is true	1
	Section-B Question No. 17 to 21 are very short answer questions carrying 2 marks each.	
17(A)	**Attempt either A or B** (I) Complete and balance the following reaction $SO_3^{2-} + MnO_4^- + H^+ \rightarrow$ (II) Why does titanium melt at a lower temperature than chromium? **OR**	1 1

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17(B)	(I) Why do titanium and scandium forms complex compounds, whereas potassium and calcium do not, although they belong to same period? (II) Complete and balance the following reaction $Fe^{2+} + Cr_2O_7^{2-} + H^+ \rightarrow$	1 1
18	"The rate of reaction remains constant during the course of reaction." Is this statement always true or is true under certain conditions? Reflect on the statement.	2
19	(I) Complete the sequence of nitrogenous bases in the given DNA strand (Figure 2) (II) Milk contains a protein called casein. How does the structure of casein change on addition of a few drops of lemon juice to milk?  Figure 2 ***(For Visually Challenged Learners)*** (I) Name the nitrogenous bases in DNA (II) Milk contains a protein called casein. How does the structure of casein change on addition of a few drops of lemon juice to milk?	1 1 1 1
20	Chlorobenzene can be converted into phenol by heating in aqueous sodium hydroxide solution at a temperature of 623K and 300 atm, however presence of a substituent group effects its activity towards this nucleophilic substitution reaction. Which substituent (i) nitro or (ii) methoxy will increase the reactivity of chlorobenzene and why?	2
21	**Account for the following:** (I) Underground iron pipelines are often protected by connecting them to a more reactive metal like magnesium. (II) Write the reactions occurring at cathode and anode during the electrolysis of molten magnesium chloride.	2x1
	Section-C Question No. 22 to 28 are short answer questions, carrying 3 marks each.	
22	(I) Which of the following has the highest magnetic moment? Justify your answer. Ti^{2+} ,Co^{2+} , Fe^{2+} . [Atomic No. Ti = 22 Co = 27 , Fe = 26] (II) Which characteristic of iron make it suitable to: a) act as catalyst in Haber's process. b) form interstitial compound with oxygen.	1 2

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23	The values of log k and 1/T for a reaction are given-	3

S.No.	Log k	1/T (/K)
1.	-4.46	0.00333
2.	-3.91	0.00323
3.	-3.37	0.00313
4.	-2.82	0.00303
5.	-2.31	0.00294

The plot of log k vs 1/T obtained for the above data is as shown in Figure 3.

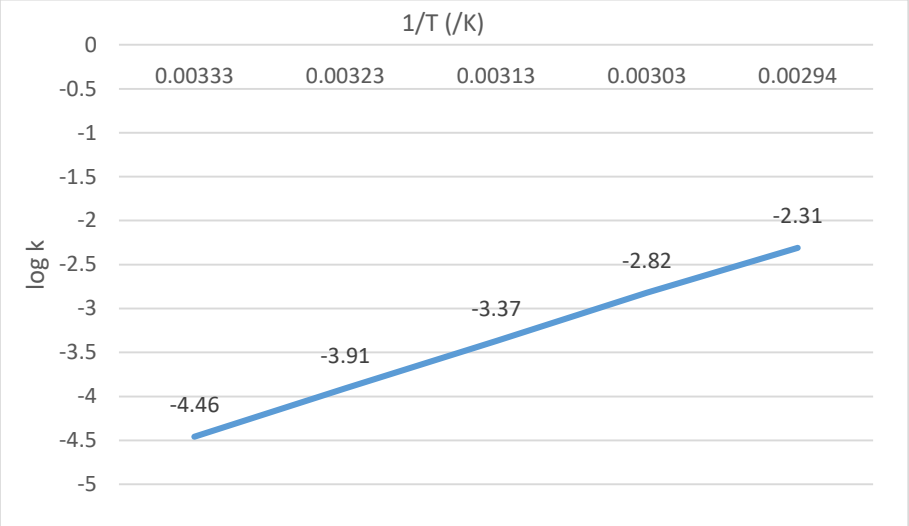


Figure 3

Here k is the rate constant and T is the absolute temperature.

Calculate E_a for the reaction. (Given R= 8.314 $JK^{-1}mol^{-1}$)

(For Visually Challenged Learners)

Rate constant k of a reaction varies with temperature 'T' according to the equation

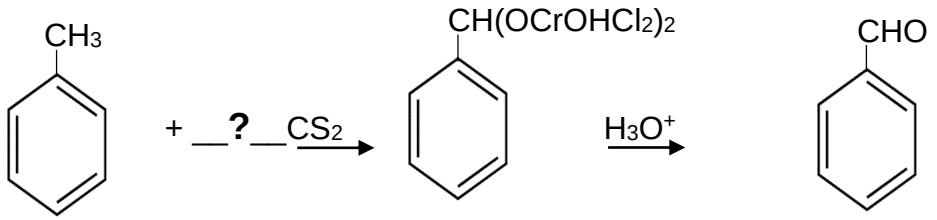
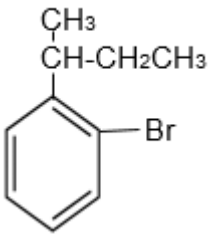
$$\log k = \log A - \frac{Ea}{2.303RT}$$

where E*a* is the activation energy. When a graph is plotted for log k vs 1/T, a straight line with a slope of − 4250 K is obtained. Calculate E*a* for the reaction. (Given R= 8.314 $JK^{-1}mol^{-1}$).

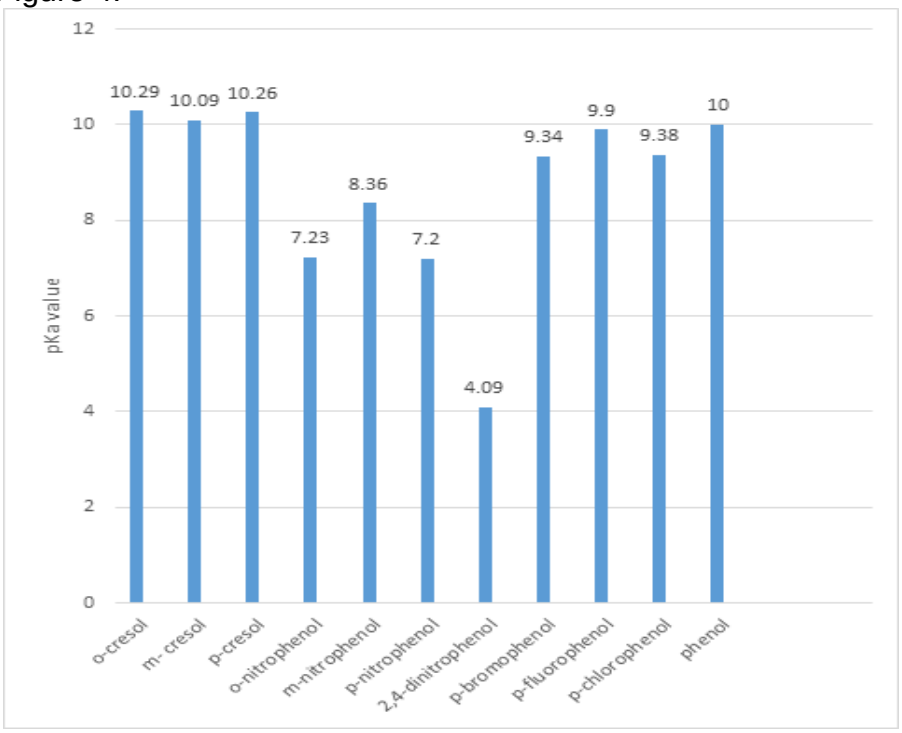
24	Identify each name reaction and complete the missing reactant/ product in the given reaction sequence.	3x1

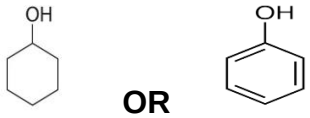
(I) $2CH_3CHO \xrightarrow{?} CH_3CH(OH)CH_2CHO$

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	(II)  (III) >C=O $\xrightarrow[-H_2O]{NH_2NH_2}$ __?_____ $\xrightarrow[\text{Heat}]{KOH/\text{ethylene glycol}}$ $>CH_2 + N_2$	
25	(I) Aman prepared a solution by dissolving 10 ml of Acetone and 10 ml of Carbon disulphide. The net volume of resulting solution was not found to be 20 mL. a) What would be the observation in this case? Comment. b) In case carbon disulphide is replaced by chloroform in the above case, what would be the expected changes in volume of the resulting solution? Support your answer with appropriate reason. (II) Calculate the concentration of oxygen (in ppm) present in 1 litre of sea water. Given that a litre of sea water weighs 1030 g and contains about 6×10^{-3} g of dissolved oxygen.	1+1 1
26	(I) Convert the following; a) Nitrobenzene to flourobenzene b) Benzoic acid to benzylamine (II) Aniline undergoes electrophilic substitution reactions readily. However, acetylation of aniline is required to obtain monosubstituted products. Give reason for your answer	2 1
	Attempt either A or B	
27(A)	(I) Write the IUPAC name of the following compound  (II) When $CH_3CH_2CH_2Cl$ reacts with AgCN and KCN, Compound "X" and "Y" respectively are formed. Identify "X" and "Y". Support your answer with appropriate reason.	1 2
	OR	
27(B)	(I) What is the IUPAC nomenclature of the compound: $(CH_3)_3C$-$C(CH_3)Br$-CH_2CH_3	1

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	(II) Which of the following will be hydrolysed more readily? $CH_2=CH-CH_2Cl$ or $(CH_3)_2CHCl$ Support your answer with reason.	2
28	(I) On adding 100 g of non-volatile solute A in 1000 g of water, the vapour gets reduced by 50 percent. Calculate the molar mass of solute A. (II) It has been observed that nitric acid solution containing 32% of water by mass, cannot be separated into its components through fractional distillation. Why?	2 1
	Section D Question No. 29 & 30 are case-based/data -based questions carrying 4 marks each.	
29	**Acidity of substituted phenols** Phenol and substituted phenols are relatively more acidic than aliphatic alcohols. The pKa value is a quantitative measure of the acidic strength. The effect of substituents on acidity of phenols (pKa values) is given in Figure 4.  **Figure 4** **Study the graph and answer the following questions** (I) The pKa value of 2,4,6- trinitrophenol will be: A. more than 8.36 B. between 7.2 and 8.3 C. between 4.09 and 2.3 D. less than 4.09 (II) The order of acidity of p-halophenols is : A. p-bromophenol > p-chlorophenol > p-flourophenol > phenol B. phenol > p-flourophenol > p-chlorophenol > p-bromophenol C. p-chlorophenol > p-bromophenol > p-flourophenol > phenol	

	D. phenol > p-flourophenol > p-bromophenol > p-chlorophenol	
	Attempt either (IIIA) or (IIIB).	
	(IIIA) Predict which one of the following two compounds will have lower value of pKa .  **OR** Give reason in support of your reason.	2
	OR	
	(IIIB) Predict whether 4 chloro-2, 6- dintirophenol is more or less acidic than 2,4,6-trinitrophenol? Use the values in the graph to support your answer.	2
	(For Visually Challenged Learners)	
	Read the passage and answer the questions that follow The presence of electron withdrawing groups such as nitro group, enhances the acidic strength of phenol. It is due to the effective delocalisation of negative charge in phenoxide ion. On the other hand, electron releasing groups, such as alkyl groups, in general, do not favour the formation of phenoxide ion resulting in decrease in acid strength. Cresols, for example, are less acidic than phenol. The pKa value is a quantitative measure of the acidic strength. Higher the pKa value lower is the acidic strength.	
	(I) Which of the following will form least stable phenoxide ion? A. phenol B. o-cresol C. o-nitrophenol D. o-chlorophenol	1
	(II) Which group causes the maximum increase in pKa value of phenol: A. – CH_3 B. – OCH_3 C. – NO_2 D. – Cl	1
	Attempt either (IIIA) or (IIIB)	
	(IIIA). Which of the following will be more acidic? Nitrophenol or 2,4-Dinitrophenol. Support your answer with appropriate reason?	2
	OR	
	(IIIB). Predict whether pka value of p-cresol or p-nitrophenol will be higher or lower than phenol. Give reason to support your answer.	2

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30	**Identification of carbohydrate** Students of class 12 are given a sample of carbohydrate obtained from a plant extract. They conducted the following tests to identify the carbohydrate. The students find that the given carbohydrate is soluble in water. They take about 1-2 mL of aqueous solution of the carbohydrate in a test tube and add 2 mL of Benedict's reagent. The test tube is heated in water bath for 3-5 minutes. A brick red precipitate is obtained. The sample of the given carbohydrate on hydrolysis with dilute acid, gives only one type of monosaccharide. Read the passage and answer the following questions: (I) The sample gives a positive Benedict's test. We can conclude that the given carbohydrate is a A. monosaccharide B. reducing sugar C. non-reducing sugar D. polysaccharide (II) Which of the following is plant based carbohydrate? (i) sucrose (ii)lactose (iii) starch (iv) glycogen A. (i) and (ii) B. (i) and (iii) C. (iii) and (iv) D. (i) and (iv) **Attempt either (IIIA) or (IIIB)** (IIIA) Considering all observations reported in the passage the sample is most likely (i) glucose (ii) fructose (iii) sucrose (iv) lactose (v) maltose. Justify your answer. **OR** (IIIB) Ravi reports that the given sample is starch. What could be his justification, based on the observations in the passage? Do you agree with Ravi? Why or why not?	1 1 2 2
Section-E Question No. 31 to 33 are long answer type questions carrying 5 marks each.		
31(A)	**Attempt either A or B** (I) Account for the following a. The pH should be around 3.5 during the addition of ammonia derivative compounds to aldehyde/ ketone. b. Alcohols are added to carbonyl compounds in the presence of dry HCl gas. c. The reaction of aldehydes with sodium hydrogen sulphite is commonly used for the separation and purification of aldehydes. (II) A student has three colourless samples: methanol, acetaldehyde, and acetic acid. They have only $NaHCO_3$, and 2,4-DNP reagent	3x1 2

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		Devise a flowchart to identify each compound using only these two reagents.	
		OR	
31(B)	(I)	Arrange the following according to the property indicated a) CH_3CHO, $HCHO$, C_6H_5CHO (decreasing order of the ease of oxidation) b) CH_3COCl, CH_3COOH, CH_3CONH_2, CH_3COOCH_3 (increasing order of nucleophilic acyl substitution) c) CH_3CHO, CH_3COCH_3, CF_3CHO (increasing order of electrophilicity of carbonyl carbon)	3x1
	(II)	Write any two possible isomers of the compound having molecular formula C_3H_6O. Identify which of these isomer/s will give a positive Tollen's test and explain the reason.	2
	Attempt either A or B		
32(A)	**Answer the following questions:**		5x1
	(I)	Two test tubes, X and Y, contain solutions of the complexes $[Cr(H_2O)_3Cl_3]$ and $[Cr(en)_3]^{3+}$ respectively. If plane-polarized light is passed through these solutions, which solution will rotate the plane of polarization and why?	
	(II)	$[Ni(CO)_4]$ and $[Ni(H_2O)_6]^{2+}$ are two complexes of Ni. Which complex will have higher crystal field splitting energy, and which factor determines this difference?	
	(III)	A solution of the coordination compound $PdCl_2 .4NH_3$ is treated with $AgNO_3$, and 2 mole of AgCl precipitate form. Determine the primary valency and secondary valency of Pd in the compound.	
	(IV)	The complex $[Fe(CN)_6]^{4-}$ exhibits d^2sp^3 hybridisation. Based on this, comment on the strength of the ligand CN^- and its effect on the geometry of the complex.	
	(V)	Why is anhydrous $CuSO_4$ white, but $CuSO_4 .5H_2O$ blue in colour?	
		OR	
32(B)	**Answer the following questions:**		5x1
	(I)	Three coordination compounds of cobalt(III) are given: Compound A: $[Co(NH_3)_6]Cl_3$ Compound B: $[Co(NH_3)_5Cl]Cl_2$ Compound C: $[Co(NH_3)_4Cl_2]Cl$ Which of the three compounds will show maximum electrolytic conductivity and why?	
	(II)	On the basis of crystal field theory, write the electronic configuration for d^5 ion if $\Delta_0 > P$.	
	(III)	Using IUPAC norms, write the formulae of Diaquadiammine (oxalato) chromium(III) chloride.	
	(IV)	Compare the optical activity of $[Co(NH_3)_2Cl_2(en)]Cl$ and $[Co(NH_3)_4Cl_2]Cl$. Which one is optically active and why?	
	(V)	Explain why $[Cu(NH_3)_4]^{2+}$ shows a blue colour while $[Cu(H_2O)_4]^{2+}$ shows a greenish colour. Which factor influences the colour difference?	

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Attempt either A or B

33(A) (I) A student investigates the effect of concentration on the cell potential of a Zn–Cu galvanic cell at 298 K. 1+2

The cell is: $Zn(s) \mid Zn^{2+}(aq) \parallel Cu^{2+}(aq) \mid Cu(s)$

standard electrode potentials:

$E^0(Zn^{2+}/Zn) = -0.76$ V, $E^0Cu^{2+}/Cu = +0.34$ V, $E^o cell = 1.10$ V

The following experimental data were collected:

Experiment	$[Zn^{2+}]$ (M)	$[Cu^{2+}]$ (M)	Measured EMF (V)
1	1.00	1.00	1.10
2	0.1	1.00	1.16
3	1.00	0.10	1.04
4	0.01	1.00	1.22
5	1.00	0.01	0.98

a) Why does experiment 4 show the maximum EMF?

b) For experiment 3, calculate the theoretical EMF using the Nernst equation. Suggest one possible reason why the measured EMF differs slightly from the calculated value.

(II) If k is expressed in Sm^{-1} and the concentration, c in mol m^{-3} then the units of molar conductivity are in Sm^2mol^{-1}. However, if we use Scm^{-1} as the units for k and $molcm^{-3}$, the units of concentration, then the units of molar conductivity are in Scm^2mol^{-1}. Show how 1 Sm^2mol^{-1} is related to 1S cm^2mol^{-1} 1

(III) What happens to the specific gravity of electrolyte in lead storage battery during charging and discharging of battery? 1

OR

33(B) I. For aqueous NaCl at 25 °C the molar conductivities at two concentrations are measured as: 3

$\Lambda_m(0.010\ M) = 123.9\ S\ cm^2\ mol^{-1}$

$\Lambda_m(0.040\ M) = 120.7\ S\ cm^2\ mol^{-1}$

Molar conductivity for strong electrolytes can be represented as $\Lambda_m = \Lambda^\circ_m - A\sqrt{c}$. Determine Λ°(molar conductivity at infinite dilution) and the constant A for NaCl solution.

II. Write the electrode reactions that occur in a hydrogen–oxygen fuel cell. Also, state two advantages of fuel cells compared to conventional batteries. 2

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SUBJECT: MATHEMATICS (041)
SAMPLE QUESTION PAPER
CLASS- XII (2026 - 27)

Maximum Marks:80 **Time Allowed: 3 Hours**

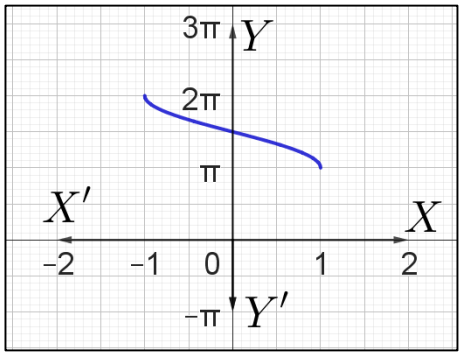
General Instructions:

Read the following instructions very carefully and strictly follow them:

1. This question paper contains 38 questions. All questions are compulsory.
2. This question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Question no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Question no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Question no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Question no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Question no. 36 to 38 are case study-based questions carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided in 2 questions in Section B, 3 questions in Section C, 2 questions in Section D and one sub-part each in 2 questions of Section E.
9. Draw neat and clean figures wherever required. Take $\pi = \frac{22}{7}$ wherever required, if not stated.
10. Use of calculators is not allowed.

SECTION – A **(20 x 1 = 20)**

This section comprises of 18 multiple choice questions and two assertion and reason type questions of 1 mark each.

Q.No.	Questions	Marks
Q1.	Identify the function whose graph is given below:  (A) $\cos^{-1}\left(\frac{x}{2}\right)$ (B) $\cos^{-1}(2x)$ (C) $\pi - \cos^{-1} x$ (D) $\pi + \cos^{-1} x$	**1**

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	For Visually Impaired: Find the range of $\frac{\pi}{2} + \sec^{-1}\left(\frac{x}{2}\right)$ (A) $[0, \pi]$ (B) $[0, \pi] - \left\{\frac{\pi}{2}\right\}$ (C) $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$ (D) $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right] - \{\pi\}$	
Q2.	Domain of the function: $\sec^{-1}(2x)$ is (A) $(-\infty, \infty)$ (B) $\left(-\frac{1}{2}, \frac{1}{2}\right)$ (C) $\left(-\infty, -\frac{1}{2}\right) \cup \left(\frac{1}{2}, \infty\right)$ (D) $\left(-\infty, -\frac{1}{2}\right] \cup \left[\frac{1}{2}, \infty\right)$	**1**
Q3.	Value of $\cot^{-1}\left(2\cos\left(\frac{1}{2}\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)\right)\right)$ (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$	**1**
Q4.	If $A = \begin{bmatrix} 2 & 1 & -2 \\ 3a & 0 & 4 \\ b & 2c & 1 \end{bmatrix}$ is a symmetric matrix, then value of $\sqrt{6a + 2b + 3c}$ is: (A) -2 (B) 2 (C) ± 2 (D) 4	**1**
Q5.	If P and Q are matrices of same order such that $P = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$, $\lvert Q \rvert = 3$, then find $\lvert adj(PQ) \rvert$ (A) -3 (B) 1 (C) 6 (D) 36	**1**
Q6.	If $\begin{bmatrix} 1 & x \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ -1 \end{bmatrix} = O$, then value of x is (A) -2 (B) -1 (C) 1 (D) 2	**1**
Q7.	The derivative of $\cos^{-1}(2x^2 - 1)$ with respect to $\cos^{-1}(x)$ is (A) 2 (B) ± 1 (C) $\frac{1}{2}$ (D) -3	**1**
Q8.	If a, b, c are arbitrary constants such that the function defined by $f(x) = \begin{cases} ax^2 + b\,, & b \neq 0\,,\ x \leq 1 \\ bx^2 + ax + c\,, & x > 1 \end{cases}$ is differentiable at $x = 1$, then which of the following could be a triplet for (a, b, c)? (A) (3,2,0) (B) (0,0,3) (C) $\left(\frac{3}{2}, 2, 0\right)$ (D) $\left(3, \frac{3}{2}, 0\right)$	**1**
Q9.	If $y = a\left\{x + 2a\ \sin\left(\frac{x}{a}\right)\right\}, a > 0, x \in [0, \pi a]$; is a decreasing function, then the interval in which x lies is (A) $\left(\frac{5\pi}{6}, \pi\right]$ (B) $\left(\frac{2\pi}{3}, \pi\right]$ (C) $\left(\frac{5\pi a}{-6}, \pi a\right]$ (D) $\left(\frac{2\pi a}{3}, \pi a\right]$	**1**

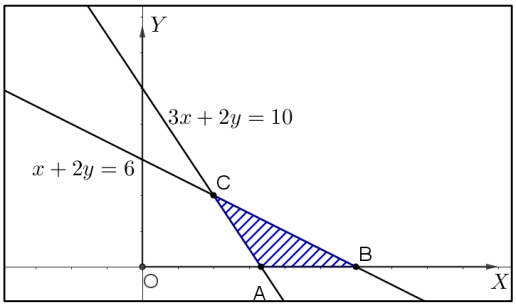
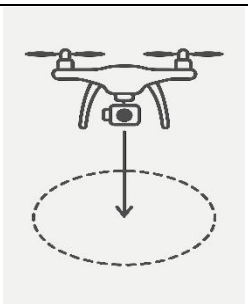
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Q10.	The values of x for which the derivative of the function $y = x^4$ is greater than the value of the function are (A) $0 < x < 4$ (B) $0 < x \leq 4$ (C) $x < 0 \; or \; x > 4$ (D) $x < 0 \; or \; x \geq 4$	**1**
Q11.	The degree and order of the differential equation $y - x\left(\frac{dy}{dx}\right)^3 = a\left(y + \frac{d^2y}{dx^2}\right)^2$ are respectively (A) $3, 1$ (B) $2, 2$ (C) $3, 2$ (D) $4, 2$	**1**
Q12.	If $\int \frac{1}{1+e^{2x}} dx = x - P \; log(1 + e^{2x}) + C$, then '$P$' is equal to (A) $-\frac{1}{2}$ (B) $-\frac{1}{3}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$	**1**
Q13.	If vector $2\hat{\imath} - u\hat{\jmath} + 3\hat{k}$ has magnitude 4, then value of 'u' is (A) 1 (B) $\sqrt{2}$ (C) $\pm\sqrt{3}$ (D) ± 2	**1**
Q14.	Angle made by the vector $\hat{\imath} + \hat{\jmath} + \sqrt{2}\,\hat{k}$ with positive direction of z-axis is (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$	**1**
Q15.	A vector perpendicular to the vectors $\hat{\imath} + \hat{\jmath}$ and $\hat{\imath} - \hat{\jmath}$ is (A) $-2\,\hat{\jmath}$ (B) $3\hat{k}$ (C) $2\hat{\imath}$ (D) $2\hat{\imath} + \hat{k}$	**1**
Q16.	The probability that it rains on a particular day is $\frac{1}{5}$. Assuming weather conditions of any two days are independent, find the probability that it rains on three consecutive days. (A) $\frac{4}{5}$ (B) $\frac{1}{5}$ (C) $\frac{1}{25}$ (D) $\frac{1}{125}$	**1**
Q17.	A die is thrown twice. What is the probability that at least one number is 4, given the sum of the numbers appearing is a multiple of 5? (A) $\frac{3}{5}$ (B) $\frac{4}{5}$ (C) $\frac{3}{7}$ (D) $\frac{4}{7}$	**1**
Q18.	A and B are two independent events such that $P(A) = \frac{1}{6}, P(B) = k$. If the probability of occurrence of exactly one of them is $\frac{1}{3}$, then find the value of k. (A) $\frac{1}{4}$ (B) $\frac{2}{3}$ (C) $\frac{3}{4}$ (D) $\frac{5}{6}$	**1**

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	Assertion-Reason Based Questions **Directions: Question numbers 19 and 20 are Assertion and Reason based questions carrying 1 mark each. Two statements are given, one labelled Assertion (A) and the other labelled Reason (R).** **Select the correct answer from the options (A), (B), (C) and (D) as given below:** (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A). (C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.	
Q19.	Assertion: The value of the integral $\int_{-1}^{1}\|x-2\|\,dx$ is 4 Reason: If $f(-x) = f(x)$ then $\int_{-a}^{a} f(x)dx = 2\int_{0}^{a} f(x)dx$	**1**
Q20.	Assertion: The general solution of a differential equation represents a family of curves. Reason: The derivative of an arbitrary constant is zero.	**1**
SECTION – B (5 x 2 = 10) **This section comprises of 5 very short answer (VSA) type questions of 2 marks each**		
Q21 (A).	Evaluate $\int \frac{1}{x\sqrt{x^6-1}}dx$	**2**
	OR	
Q21 (B).	Evaluate $\int \frac{\cos x-\sin x}{\sqrt{\sin(2x)}}\,dx$	
Q22.	If $y = \sin(x+y)$ and $\frac{dy}{dx} = 0$, then prove that $y = \pm 1$.	**2**
Q23 (A).	A line passing through the points $(2,6,5), and\ (3,p,7)$ is parallel to the line joining the points $(-4,3,4)$ and $(-1,0,q)$. Find the value of $\frac{q}{p}$.	**2**
	OR	
Q23 (B).	A rectangle ABCD is such that $\overrightarrow{AC} = \vec{u}, and\ \overrightarrow{BD} = \vec{v}$, then find the vectors representing four sides of the rectangle (in a cyclic order starting from $\overrightarrow{AB}$).	
Q24.	For two non-zero vectors $\vec{a}$ and $\vec{b}$, $\|\vec{a}\| = \sqrt{6}$, $\vec{a}\cdot\vec{b} = 1$ and $\vec{b}\times\vec{a} = 3\hat{\imath} + 4\hat{\jmath} - 2\hat{k}$. Find $\|\vec{b}\|$.	**2**

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Q25.	The feasible region of a linear programming problem is shown in the given figure:  Write all the constraints on the decision variables involved **For Visually Impaired:** For a linear programming problem, the maximum value of an objective function $Z = 3x + 4y$ exists at the points (p, q) and $\left(2p, \frac{q}{2}\right)$. Find the relation between p and q.	**2**
	SECTION – C (6 x 3 = 18) **This section comprises of 6 short answer (SA) type questions of 3 marks each.**	
Q26.	For two matrices, $P = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 2 \end{bmatrix}$ and $Q = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 1 & 0 \end{bmatrix}$, find PQ and QP, whichever is defined. Also, find the inverse of the product matrix, if it exists.	**3**
Q27.	The circular area of vision on the ground by a surveillance drone fitted with camera increases at the rate of $44\ m^2$ per metre as it gains vertical height. The radius of the circle on the ground increases at the rate of 6 m/s. Find the rate at which the height of the drone increases with time, at the instant when the radius of the circular area on the ground is 14 m. $\left(Use\ \pi = \frac{22}{7}\right)$ 	**3**
Q28 (A).	If $x = sin^3 t, y = cos^3 t$, $Evaluate\ \frac{d^2y}{dx^2}\ at\ \left(\frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}}\right)$.	**3**
	OR	
Q28 (B).	If $y = x^{logx} + (logx)^x$; $x > 1$, then evaluate $\frac{dy}{dx}$	
Q29 (A).	Find the particular solution of the differential equation $xdy + (y - x^2e^x)dx = 0$, given that y (1) =0	**3**
	OR	
Q29 (B).	Find the particular solution of the differential equation $(x^2 + y^2)dy = xydx$, given that $y(0) = 1$	

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Q30 (A).	Find the foot of the perpendicular drawn from the point (2,1,4) on the line $x-2=\frac{2y+3}{4}=2-z$. Also find the equation of the line passing through this point and the point (2,1,4).	**3**
	OR	
Q30 (B).	Find the angle between the lines: $\frac{x-1}{2}=\frac{y-1}{2}=\frac{z-2}{1}$ and $\frac{x-5}{2}=y-\frac{1}{2}=\frac{3-z}{2}$. Also find the cartesian equation of a line passing through the point (1,1,1) and perpendicular to both lines.	
Q31.	Solve the linear programming problem graphically: Maximize $Z=40x+30y$ Subject to constraints: $x+y\leq 6, 2x\geq y, y\geq 2$ **For Visually Impaired:** For a linear programming problem, vertices of the feasible region are $(20,0),(40,0),(30,10),(0,20)$ and $(0,10)$. Minimize and maximize the objective function $Z=50x-20y$	**3**
	SECTION – D (4 x 5 = 20) **This section comprises of 4 long answer (LA) type questions of 5 marks each**.	
Q32 (A).	Let $A=N\times N$, where N is the set of natural numbers and a relation R is defined on A such that $(a,b)R(c,d)\ if\ and\ only\ if\ a^2+d^2=b^2+c^2$. Check whether R is an equivalence relation or not?	**5**
	OR	
Q32 (B).	A relation $R:(-\infty,\infty)\to[0,\infty)$ is defined as $R=\{(x,y):y=x^2+5\}$ Draw its graph and check whether it is a function or not. If the given relation is a function, then examine whether the function is injective, and whether it is surjective or not. **For Visually Impaired:** A relation $R:(-\infty,\infty)\to[0,\infty)$ is defined as $R=\{(x,y):y=x^2+7\}$. Check whether it is a function or not. If the given relation is a function, then examine whether the function is injective, surjective, or both, and conclude whether it is bijective or not.	
Q33.	Evaluate $\int_{-a}^{a}\sqrt{\frac{a-x}{a+x}}\,dx$	**5**
Q34 (A).	Using Integration, find the area of the region $\{(x,y):0\leq y\leq\sqrt{4-x^2}, x\geq 2-y\}$.	**5**
	OR	
Q34 (B).	Using Integration, find the area of the region bounded by the curve $y^2=2x$ and the line $x-2y=0$.	

*There is no change in the Question Paper Design and Assessment Pattern for Academic Session 2026-27

	For Visually Impaired:	
Q34 (A).	Using Integration, find the area of the region enclosed within $25x^2 + 16y^2 = 400$	**5**
	OR	
Q34 (B).	Using Integration find the area of the region in the first quadrant bounded by the curve $y^2 = 8x$, and the lines $x = 2, x = 5$ and x-axis.	
Q35.	A fitness app syncs with four types of smartwatches: Alpha, Beta, Gamma and Delta. 45% of the users use Alpha, 30% use Beta,15% use Gamma and 10% use Delta. Due to device errors, the probabilities of getting a false heart-rate alert from the Alpha, Beta, Gamma and Delta smartwatches are respectively 0.01, 0.02, 0.05, and 0.01. Based on the given information, answer the following questions: (i) Find the probability that a randomly selected user gets a false alert. (ii) If a false alert occurs, what is the probability that it came from a Beta smartwatch?	**5**

SECTION – E (3 x 4 = 12)

This section comprises of 3 case-study/passage-based questions of 4 marks each with sub parts. The first two case study questions have three sub parts (i), (ii), (iii) of marks 1, 1, 2 respectively. The third case study question has two sub parts of 2 marks each.

Q36. Two security sensors in a multi-storey mall emit laser beams for movement detection. Three of these beams travel along the following lines:

$l_1: \vec{r} = (\lambda + 1)\hat{\imath} + (3\lambda + 2)\hat{\jmath} + (3 - \lambda)\hat{k}$ and

$l_2: \vec{r} = 2(\mu + 1)\hat{\imath} + (3\mu - 1)\hat{\jmath} + (\mu + 1)\,\hat{k}$

$l_3: \vec{r} = (\gamma + 3)\hat{\imath} + (3\gamma - 1)\hat{\jmath} + (1 - \gamma)\,\hat{k}$



Based on the given information, answer the following questions:

(i) Show that the lines l_1 and l_2 are not parallel to each other. **1**

(ii) Show that the lines l_1 and l_3 are parallel to each other. **1**

(iii) (a) Find the shortest distance between the lines l_1 and l_2.

OR

(b) Find the shortest distance between the lines l_1 and l_3. **2**

*There is no change in the Question Paper Design and Assessment Pattern for Academic Session 2026-27

Q37.	Water is essential for life, and rivers are vital water sources across the world. The Yamuna Action Plan (YAP), one of India's largest river restoration projects, was launched to clean the Yamuna, but its effectiveness (WQI) the Water Quality Index must be measured through cost-benefit, economic and environmental analysis. Therefore, comprehensive pollutant treatment, prevention of further contamination, and community awareness are crucial for sustainable river management. Implementing the same concept to the river Yamuna, if an initial base line WQI is 35, a quadratic expression is formed to measure the impact of investment of ₹x crore on the WQI i.e., $W(x)$ as $W(x) = 35 + 1.6x - 0.02x^2$, where x is in Rupees in crores  Use the given $W(x)$ to answer the following questions: (i) Find the value of x at the stationary point for the function $W(x)$. (ii) Find the maximum value of the WQI using the second derivative test? (iii) (a) If the investment is ₹ 35 crores, find by how much the WQI is short of its maximum achievable value. **Or** (b) Evaluate the WQI when the investment is ₹60 crore. Determine whether this indicates an enhancement or a diminishing return in our objective when compared with the optimum WQI.	**1** **1** **2**
Q38.	To celebrate their child's birthday, a family decided to distribute fruits among children in three orphanages. The fruit vendor packed three different types of baskets, according to the quantity required at each orphanage, details are given as follows: Basket A contains 5 kg Apples, 10 kg Oranges and 5 kg Pomegranate Basket B contains 10 kg Apples, 10 kg Oranges and 5kg Pomegranate Basket C contains 8 kg Apples, 15 kg Oranges and 7kg Pomegranate Cost of basket A, B and C are respectively ₹2750, ₹3500, and ₹4100.  Based on the given information, answer the following questions: (i) Write the matrix form of the equations involved in this situation and find the adjoint of the coefficient matrix involved. (ii) Using inverse of the matrix, calculate the price of each fruit per kg.	**2** **2**

*There is no change in the Question Paper Design and Assessment Pattern for Academic Session 2026-27

BIOLOGY – CODE NO. 044
SAMPLE QUESTION PAPER*
CLASS – XII (2026-27)

Maximum Marks: 70 **Time: 3 hours**

General Instructions:

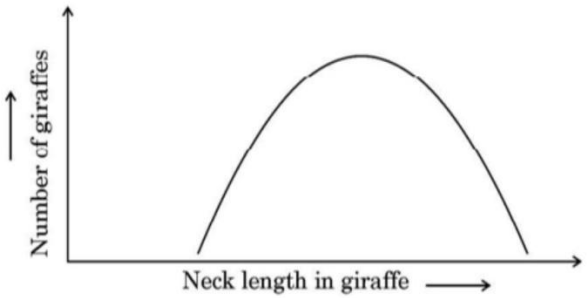
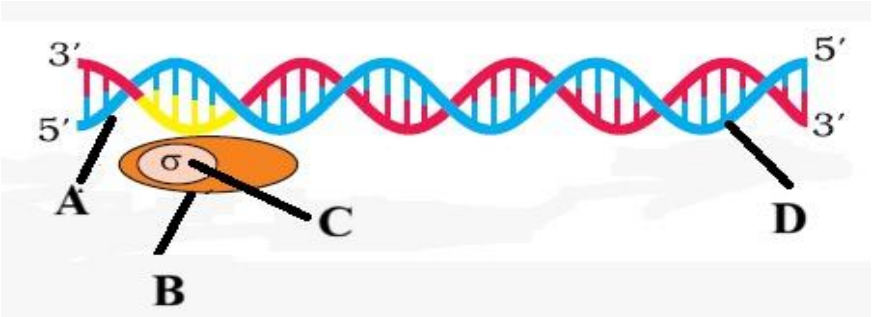
(i) All questions are compulsory.
(ii) The question paper has five sections and 33 questions.
(iii) Section–A has 16 questions of 1 mark each; Section–B has 5 questions of 2 marks each; Section– C has 7 questions of 3 marks each; Section– D has 2 case-based questions of 4 marks each; and Section–E has 3 questions of 5 marks each.
(iv) There is no overall choice. Answer all 33 questions. However, internal choices have been provided in some questions. A student has to attempt only one of the alternatives in such questions.
(v) Wherever necessary, neat and properly labeled diagrams should be drawn.

Section – A

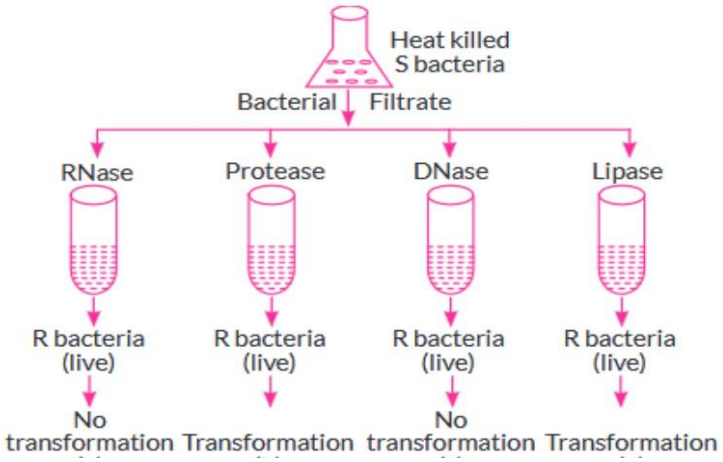
Q. Nos. 1 to 12 are multiple choice questions. Only one of the choices is correct. Select and write the correct choice as well as the answer to these questions.

Q. No.	Question	Marks
1	A researcher observes that in a particular flowering plant, the pistil matures several days before the stamens in all flowers of the plant. What is the most significant consequence of this timing difference? A. It guarantees self-pollination B. It promotes autogamy C. It ensures cross-pollination D. It leads to geitonogamy	1
2	A male patient undergoes a medical procedure where a small section of the vas deferens is cut and tied at both ends. Which of the following effects is the intended and direct result of this procedure? A. Blockage of sperm transport into the ejaculatory duct and urethra. B. Inhibition of testosterone production by the testes. C. Suppression of the formation of seminal plasma by accessory glands. D. Prevention of spermatogenesis in the seminiferous tubules.	1
3	During the early stage of human embryonic development, the blastomeres are arranged into an outer layer and an inner group of cells known as the Inner Cell Mass (ICM). What is the fundamental significance of the outer layer, the trophoblast? A. It is responsible for forming the primary germ layers (ectoderm, mesoderm, endoderm).	1

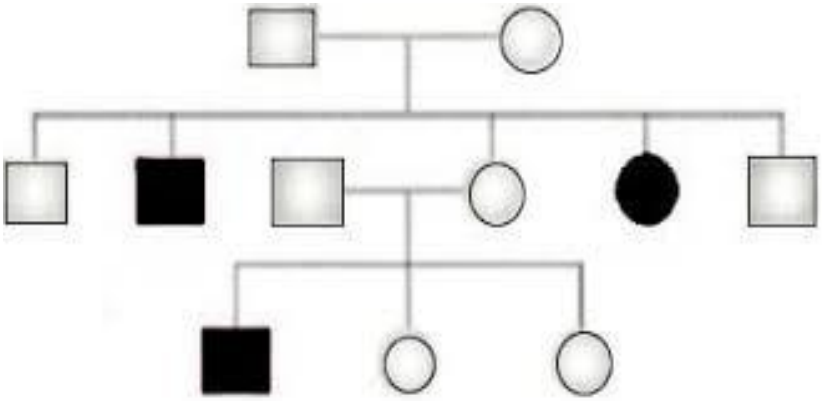
**There is no change in the Question Paper Design and Assessment Pattern for Academic Session 2026-27.*

	B. It provides essential nutrients to developing embryo and helps in its implantation. C. It secretes colostrum which contains antibodies. D. It gives rise to all the organs of the future embryo (organogenesis).	
4	Select the option that gives the correct description of the process of natural selection with respect to the length of the neck of the giraffe.  A. Stabilising selection as giraffes with longer neck lengths are selected further. B. Disruptive selection as giraffes with smaller and longer neck lengths are selected. C. Directional selection as giraffes with longer neck lengths are selected. D. Stabilising selection as giraffes with medium neck lengths are selected. - - - - - - - - - - - - - - - - - - - *For Visually impaired students* Evolution of antibiotic-resistant bacterial population signifies the information that A. Acquired traits are inherited. B. Nature selects for fitness. C. Genetic variations are not a prerequisite factor for natural selection. D. The theory of spontaneous generation of life holds true.	1
5	The initiation step during the process of transcription in bacteria is shown below. Identify A, B, C and D by selecting the option:  A. A-RNA polymerase, B-Promoter, C-Sigma factor, D-DNA helix B. A-Promoter, B-Sigma factor, C-RNA polymerase, D-DNA helix C. A-Promoter, B-RNA polymerase, C-DNA helix, D-Sigma factor D. A-Promoter, B-RNA polymerase, C-Sigma factor, D-DNA helix	1

**There is no change in the Question Paper Design and Assessment Pattern for Academic Session 2026-27.*

	- - - - - - - - - - - - - - - - - - *For Visually impaired students* The promoter site and the terminator site for transcription are located at: A. 3′ (downstream) end and 5′ (upstream) end, respectively of the transcription unit B. 5′ (upstream) end and 3′ (downstream) end, respectively of the transcription unit C. the 5′ (upstream) end D. the 3′ (downstream) end	
6	Given below is the illustration of the different steps of experiments conducted by MacLeod, McCarty and Avery to find the chemical nature of the 'transforming principle' as DNA. Select the option that **incorrectly** depicts the step of the experiment.  A B C D - - - - - - - - - - - - - - - - - - *For Visually impaired students* In the Griffith experiment, which of the following, when injected into mice, did not cause pneumonia? A. Heat-killed S strain only B. Live R strain only C. Both (A) and (B) D. Live S strain only	1
7	A scientist was culturing *E.coli* in a $^{15}NH_4Cl$ medium initially, then he shifted *E.coli* to a $^{14}NH_4Cl$ medium. The DNA was extracted from *E.coli* after 60 minutes of transfer to $^{14}NH_4Cl$ Medium and performed density gradient centrifugation to check the isotope distribution in newly replicated DNA. What will be the densities of DNA molecules formed? A. 25% hybrid and 75% light B. 50% hybrid and 50% light C. 75% hybrid and 25% light D. 85% hybrid and 15% light	1

**There is no change in the Question Paper Design and Assessment Pattern for Academic Session 2026-27.*

8	Study the given pedigree and identify the disease represented  A. Myotonic Dystrophy B. Sickle cell anaemia C. Haemophilia D. Colour blindness - - - - - - - - - - - - - - - - - - - *For Visually impaired students* If a genetic disease is transferred from a phenotypically normal but carrier female to only some of the male progeny the disease is: A. Autosomal dominant B. Autosomal recessive C. Sex-linked dominant D. Sex-linked recessive	1
9	A patient with fever, chest pain, cough and breathing difficulty undergoes a chest X-ray. The film shows patchy opacities and reduced clarity of lung fields. Which observation best supports pneumonia? A. Enlarged heart size B. Inflammation or fluid buildup in the lungs C. Fractured ribs D. Air pockets in the lungs	1
10	A stretch of an euchromatin has 200 nucleosomes, how many base pairs (bp) will be there in the stretch and what would be the length of the typical euchromatin? A. 20,000 bp and $13{,}000 \times 10^{-9}$ m B. 10,000 bp and $10{,}000 \times 10^{-9}$ m C. 40,000 bp and $13{,}600 \times 10^{-9}$ m D. 40,000 bp and $13{,}900 \times 10^{-9}$ m	1
11	A biotechnology company grew Streptomyces in a fermenter to produce an antibiotic. It was observed that Day 7 onwards, the actual yield drops sharply, even though nutrient level, temperature, and aeration remain	1

	unchanged. Also, Microscopy revealed new colonies of bacteria which were not present in the original culture. So the fermenter was stopped. What measures should be taken to prevent the fall in yield in future fermentations? A. Increasing temperature only at the end of fermentation to kill contaminants. B. Adding antibiotics to the culture so unwanted microbes cannot grow. C. Ensuring sterile handling, sterilised equipment, and aseptic transfer before and during fermentation. D. Using a genetically modified strain designed to grow faster than contaminants.	
12	A rice farmer's field, which typically thrives in waterlogged conditions and requires soil rich in nutrients like phosphorus, is experiencing a severe drought this year. The crop is also susceptible to root infestation by pathogens. Which of the following techniques is best to address all these issues? A. Adding nitrogen-fixing bacteria to the soil B. Applying chemical fertilizers C. Adding spores of *Glomus* to the soil D. Flooding the field artificially to mimic normal conditions	1

Question No. 13 to 16 consist of two statements – **Assertion** (A) and **Reason** (R). Answer these questions selecting the appropriate option given below:

A. Both A and R are true and R is the correct explanation of A.
B. Both A and R are true but R is not the correct explanation of A.
C. A is true but R is false.
D. A is False but R is true.

13	**Assertion** (A): Spermatogenesis is an ongoing process throughout the reproductive life of a human male, while oogenesis is a discontinuous process. **Reason** (R): Oogenesis begins during the embryonic stage, gets arrested at prophase-I, resumes after puberty, and is arrested again at metaphase-II.	1
14	**Assertion** (A)**:** There is no variation in the percentage recombination due to loosely or tightly linked genes. **Reason** (R): Linked genes result in occurrence of higher proportion of parental gene combinations.	1
15	**Assertion** (A)**:** The primary goal of sewage treatment is to eliminate all organic matter from the effluent. **Reason** (R)**:** High BOD indicates a low level of organic matter in the water, which can lead to increased oxygen consumption by microorganisms, potentially harming aquatic life.	1
16	**Assertion** (A)**:** Tissue culture is used when traditional breeding technique is insufficient to keep pace with the demand of crop improvement.	1

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	Reason (R)**:** Totipotency is the ability of a single plant cell or explant to regenerate into a whole plant under sterile conditions in nutrient media.	
	Section - B	
17	***Attempt either option A or B.*** A. Geitonogamy can be compared with cross-pollination as well as self-pollination. Comment. **OR** B. Mohit wants to produce hybrid seeds in a plant in his field in which male and female flowers mature at the same time and are close together. Suggest one strategy to promote only desired cross pollination and justify your answer.	2
18	The movement of landmasses over geological time changed the distribution and evolution of animals. Explain this idea using two examples from different regions of the world.	2
19	The following is a realistic example dataset and not a real world patient dataset: **Sample ID** | **Tumor size (mm^3)** | **Glucose level mM/litre** | **Oxygen level (mmHg)** | **Neighboring cell's viability/ vitality (%)** 1 | 150 | 2.0 | 15 | 60 2 | 50 | 4.5 | 30 | 85 3 | 200 | 1.3 | 10 | 50 4 | 100 | 3.0 | 20 | 70 5 | 250 | 0.9 | 7 | 40 A. Interpret the relationship between tumor size and glucose level. B. Evaluate the data to determine how tumor progression may affect the viability/vitality of neighboring cells. Provide a brief explanation.	2
20	***Attempt either option A or B.*** A. The table below shows hypothetical data comparing α-lactalbumin concentration in three types of milk, including that of Rosie, a transgenic cow designed to express human milk proteins. **Source of Milk** | **α-lactalbumin Content approximately (g/L)** Normal cow's milk | 0.1 Rosie cow's milk | 2.4 Human milk | 3.5	2

**There is no change in the Question Paper Design and Assessment Pattern for Academic Session 2026-27.*

I. Assuming all other nutritional components are identical, which milk would provide the lowest nutritional contribution from α-lactalbumin?.

II. If further genetic modification increases α-lactalbumin levels in Rosie to 3.5 g/L, predict one ethical concern that might arise from such a development.

OR

B. A farmer replaces the GM crop with a conventional variety. Predict any two challenges she/he may face. Support your answer with two reasons.

21 ***Attempt either option A or B.*** 2

A. The table below is a hypothetical data on Seasonal Temperatures seen over two different zones

Region	Temp. in Spring (^oC)	Temp. in Summer (^oC)	Temp. in Autumn (^oC)	Temp. in Winter (^oC)	Primary Productivity (GPP) (g/m^2/ year)	Respiration (R) (g/m^2/year)
Region X	27	28	27	26	3500	1300
Region Y	12	20	14	4	2000	800

I. Justify which region will have greater species diversity based on the given data?

II. Calculate the net primary productivity (NPP) of the region mentioned in (I) and give its significance?

OR

B. Following is a hypothetical data on Plant Pollinator interaction

Plant	Main Pollinator	Seed set (%) with pollinator	Seed set (%) without pollinator
P1	H1 (Bee)	80	40
P2	H2 (Butterfly)	75	45
P3	H3 (Moth)	85	35

I. From the data above, justify which plant shows the highest dependence on its main pollinator?

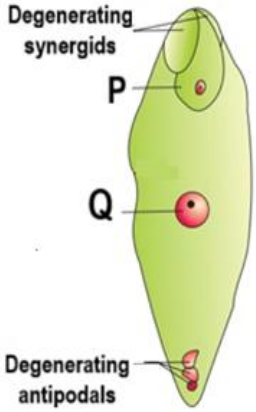
II. What kind of ecological interaction is shown in the table above? Define it in one sentence.

Section - C

22 A couple, A and B, visit a fertility clinic. Examination reveals that female A has a blockage in her fallopian tubes, but her ovaries can produce viable ova. Male B produces sperm with a very low sperm count, making natural fertilization unlikely. 3

Which two different Assisted Reproductive Technologies (ARTs) would a doctor

**There is no change in the Question Paper Design and Assessment Pattern for Academic Session 2026-27.*

	most likely recommend for this couple to increase their chances of conceiving? Justify your selection by briefly stating the principle behind each chosen technique concerning the couple's specific issues.	
23	In the post-fertilization embryo sac shown in figure, structure 'P' results from the fusion of one male gamete with the egg cell, and structure 'Q' results from the fusion of the other male gamete with the polar nuclei.  Figure: Fertilised embryo sac in a flowering plant A. Identify structure 'P' and structure 'Q' and state their ploidy. B. Evaluate the following statement: "The division of 'P' always precedes the division of 'Q'." Justify your evaluation based on the nutritional requirements for seed development. C. Which characteristic feature of endosperm helps in its nutritive role? - - - - - - - - - - - - - - - - - - - *For Visually impaired students* A. Which two products are formed after double fertilization in the embryo sac of a flowering plant? B. State the ploidy of these two structures. C. Which of these two structures develops first during embryonic development? State its significance.	3
24	In mice, the traits for running ability and hair colour assort independently. Running (R) is dominant over walking in circles (r). Black hair (B) is dominant over brown hair (b). A heterozygous running, brown-haired female mouse was artificially inseminated using sperm from a heterozygous running, heterozygous black male. A. Write the genotypes of both parents and draw a Punnett square showing all possible offspring combinations. (1) B. Using the Punnett square, state the phenotypic ratio for: Running–Black, Running–Brown, Walking–Black, Walking–Brown. (1) C. If 64 offspring were produced, calculate how many would be expected to show the Running–Black phenotype. (1)	3

**There is no change in the Question Paper Design and Assessment Pattern for Academic Session 2026-27.*

25	How can a single gene defect caused by the mutation result in Phenylketonuria (PKU)? Explain the molecular mechanism underlying pleiotropy.	3
26	Drug-use disorders often begin with occasional consumption but may progress into a compulsive state. Analyse how addiction differs from dependence in both origin and consequence, and justify why long-term drug intake forces the user to increase dose over time.	3
27	Pathogens are often considered harmful, yet they have been adapted as powerful tools in genetic engineering. Explain how pathogens have been modified into useful vectors. Describe the role of a similarly modified plant vector and animal vector.	3
28	The nematode *Myrmeconema neotropicum* lives inside the abdomen of certain tropical ants (*Cephalotes atratus*). Inside the ant's body, the nematode develops slowly without killing it. Once mature, it changes the color of the ant's abdomen to bright red, making it resemble a berry. Birds, mistaking it for a fruit, eat the ant. The nematode's eggs survive in the bird's digestive system and pass out in droppings, which other ants contact while foraging, starting the cycle again. A. Why would harming the ant too early be disadvantageous for the nematode? (1) B. How is natural selection playing a vital role in maintaining this interaction? (1) C. What kind of ecological interaction is seen between nematode and ant? Define it in one sentence. (1)	3
	Section - D	
29	Madhuri, a 17-year-old girl, has been experiencing irregular periods for the past one year. Sometimes, her cycles are as short as 20 days and sometimes as long as 45 days. She occasionally has very heavy bleeding that lasts over a week, causing fatigue and weakness. A doctor reviews her menstrual diary and explains that these changes can be linked to hormonal fluctuations in the hypothalamic-pituitary-ovarian (HPO) axis. He shows Madhuri the lack of a clear ovulation peak and low progesterone levels in her recent cycles. A. Madhuri experiences heavy bleeding lasting over a week causing fatigue. What possible complication could arise from this condition? (1) B. Examine how the absence of a distinct LH (luteinizing hormone) surge in the menstrual cycle might explain both the irregularity and heaviness of Madhuri's periods. (2) *Attempt either subpart C or D.* C. Madhuri's cycles vary between 20 and 45 days. What might this irregularity indicate about her ovarian function? (1)	4

**There is no change in the Question Paper Design and Assessment Pattern for Academic Session 2026-27.*

	OR D. If a doctor suspects hypothalamic-pituitary dysfunction in Madhuri, which hormone secretion pattern would you expect to be disrupted? (1)	
30	A tribal farmer is cultivating *Brassica oleracea* (cabbage) in a forest region rich in pollinators and rare butterflies. Recently, his crops have been heavily damaged by cabbage looper caterpillars. To manage the pest outbreak, he applies Nucleopolyhedrovirus (NPV). After a few weeks, the pest population decreases significantly without affecting other insects or animals in the area. A. Justify with reasons why the use of Nucleopolyhedrovirus (NPV) was a suitable choice for controlling cabbage looper caterpillars. (1) B. Suggest another IPM strategy suitable for controlling caterpillar pests, and briefly describe its mode of action. (2) *Attempt either subpart C or D.* C. Name the group of pathogens to which *Nucleopolyhedrovirus* belong. (1) **OR** D. Justify why the use of NPV is considered a biocontrol method. (1)	4
	Section - E	
31	***Attempt either option A or B.*** A. A short stretch of DNA strand that codes for a polypeptide is shown below: 3'—TACAAAAAGAAGAAAAAGAAGATT--5' Due to exposure to harmful UV radiation two different types of errors occurred during DNA replication affecting their sequences of bases in the DNA in two different cells. I. What is the term given to a segment of DNA that codes for a polypeptide? II. Which type of mutation could have occurred in each of the following due to the error during replication? a) 3'-- TACAAAAAGAAGAAAAAGAAAATT--5' b) 3'-- TACAAAAAGAAGAAAAAGAGATT--5' III. If DNA strand (a) codes for Methionine and sequences of Phenylalanine, mention the sequence of amino acids that will constitute the polypeptide chain after translation. How many amino acids will be translated from this strand? IV. If all the codons of the stretch code for Phenylalanine excluding the initiator codon which feature does it depict? **OR** B. I. If GUA is an intron how many amino acids will be translated from the mature or processed RNA. Pick out the untranslated region from the following messenger RNA and mention their location and sequence.	5

**There is no change in the Question Paper Design and Assessment Pattern for Academic Session 2026-27.*

	What is its significance? 5'ACGUCGAUGACCGUAGCGUUUGUAUCUUUAGUAGUGGUAUUA GUAGGCUAAAAA3' II. If AAA is the anticodon of phenylalanine, draw the tRNA adaptor molecule, Mention the position of amino acid Phenylalanine in the sequence of polypeptide counting the triplet codons.	
32	***Attempt either option A or B.*** A. A scientist is screening bacterial transformants engineered with DNA fragments of the β-globin gene. A radioactive DNA probe complementary to the normal β-globin sequence is used. After autoradiography: • Plates A and C show strong and week radioactive signals respectively. • Plate B shows no radioactivity. Answer the following: I. Why does plate B fail to show radioactivity, while A and C do? (2) II. How can this experiment help in distinguishing a normal individual, a carrier, and a patient of sickle cell anaemia? (2) III. Name one other commonly used technique (apart from autoradiography) to differentiate recombinants from non-recombinants. (1) **OR** B. A student placed DNA from three different sources: bacteria, plant tissue, and fungal cells in a test tube A, B and C respectively. During the process of isolating and purifying DNA, the student added cellulase to all test tubes. I. Predict and explain in which trial(s) DNA isolation would fail, and why? Also suggest how it could be corrected. (2) II. After rectifying the mistake made in 'a', outline the steps to obtain visible spooled DNA from the sample. (2) III. Suggest one application of the spooled DNA in biotechnology. (1)	5
33	***Attempt either option A or B.*** A. A Green Valley *X*, home to rare medicinal plants and several migratory bird species, is facing multiple environmental challenges. I. A luxury brand of cosmetics with little regulation has been sourcing wild medicinal plants from the valley. II. The company has introduced an ornamental grass species from another country, which has taken over large areas of native grassland. III. There is also a reported drastic decline in the endemic trees. IV. Large scale forest fire in a part of the valley. a) From the above hypothetical case, identify and explain the four major biodiversity threats. (4) b) Suggest one in-situ conservation practice that could help protect the biodiversity of Green Valley *X*, and justify your choice. (1)	5

**There is no change in the Question Paper Design and Assessment Pattern for Academic Session 2026-27.*

	OR B. At a composting ground, after a community clean-up drive, large amounts of vegetable peels, garden leaves, and other organic waste are piled up. Over the next few weeks, the volume of the waste decreases noticeably. I. Explain the sequence of processes that could have led to this transformation, and name the major organisms involved at each step. (4) II. If a prolonged dry spell occurs during this composting period, explain how it could affect the rate of the above processes. (1)	

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**There is no change in the Question Paper Design and Assessment Pattern for Academic Session 2026-27.*

SAMPLE QUESTION PAPER
Class XII Session 2026-27
COMPUTER SCIENCE (Code 083)

TIME: 3 hours **MAX. MARKS: 70**

General Instructions:

- This question paper contains 37 questions.
- All questions are compulsory. However, internal choices have been provided in some questions. Attempt only one of the choices in such questions
- The paper is divided into 5 Sections- A, B, C, D and E.
- Section A consists of 21 questions (1 to 21). Each question carries 1 Mark.
- Section B consists of 7 questions (22 to 28). Each question carries 2 Marks.
- Section C consists of 3 questions (29 to 31). Each question carries 3 Marks.
- Section D consists of 4 questions (32 to 35). Each question carries 4 Marks.
- Section E consists of 2 questions (36 to 37). Each question carries 5 Marks.
- All programming questions are to be answered using Python Language only.
- In-case of MCQ, text of the correct answer should also be written.

Q No.	Section-A (21 x 1 = 21 Marks)	Marks
1	Write the output on execution of the following Python code: L1=[10,20,25,30] L1.insert(1,15) print(L1)	1
2	What will be the output of the following expression? Msg = "programming is amazing" print(Msg.split("m")) a) ['progra', '', 'ing is a', 'azing'] b) ['progra', 'ming is a', 'azing'] c) ('progra', '', 'ing is amazing') d) ['program', 'ing is a', 'zing']	1
3	Which of the following data types in Python is mutable? a) Tuple b) String c) Integer d) List	1
4	A column in one table matching a primary key of another table is called a: a) Composite key b) Candidate key c) Foreign key d) Super key	1
5	Consider the following Python code: t = (1, 2, 3) l = [4, 5] print(t + l) What will happen when this code is executed? a) [1, 2, 3, 4, 5] b) (1, 2, 3, 4, 5) c) Error, because a tuple cannot be concatenated with a list	1

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	d) ((1, 2, 3), [4, 5])	

6	What possible output is expected to be displayed on the screen at the time of execution of the Python program from the following code?
	import random
	games = {0: "Chess",1: "Scrabble",2: "Sudoku",3: "Tic-Tac-Toe",4: "Puzzle"
	}
	start_key = random.randint(1, 2)
	end_key = random.randrange(3, 4)
	for k in range(start_key, end_key + 1):
	print(games[k], end="@@")
	a) Sudoku@@Tic-Tac-Toe@@Puzzle@@
	b) Scrabble@@Sudoku@@Tic-Tac-Toe@@
	c) Chess@@Scrabble@@Tic-Tac-Toe@@
	d) Scrabble@@Tic-Tac-Toe@@
		1

7	Predict the output for the following: Dl={"Pen":12, "Pencil":6, "Scale":5, "InkPen":20} print(12 in Dl) a) False b) Error c) None d) True	1
8	Long-distance communication without line-of-sight is mainly achieved using: a) Microwaves b) Ultraviolet c) Radio waves d) X-rays	1
9	Which MySQL query will display all employees whose salaries are between 90,000 and 100,000 (including limits)? a) SELECT * FROM Employees WHERE salary > 90000 AND salary < 100000; b) SELECT * FROM Employees WHERE salary IN (90000, 100000); c) SELECT * FROM Employees WHERE salary BETWEEN 90000 AND 100000; d) SELECT * FROM Employees WHERE salary = 90000 OR salary = 100000;	1
10	What will be the output of the following Python statement: print(25+100**1**3/5)	1
11	What is the data type of the result when the expression 5 // 2 is evaluated in Python? a) float b) int c) string d) None	1
12	Every network interface card (NIC) comes with its own _____ address. a) Internet Protocol (IP) b) Open Systems Interconnection (OSI) c) Physical (MAC) d) Dynamic Host Configuration Protocol (DHCP)	1
13	In a student table, “Class” can only take values from {IX, X, XI, XII}. This set is called the _____. a) Domain b) Tuple	1

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	c) Attribute d) Key	

14	What would be written in the file errorlog.txt after the following code is run?
	def data(num):
	try:
	return 190/(29-num)
	except ZeroDivisionError:
	logfile = open ("errorlog.txt", "w")
	logfile.write ("Num was 29")
	logfile.close()
	data(29)
		1

15	Which of the following will NOT change if you insert a new record into a table? a) Cardinality b) Degree c) Number of tuples d) Total rows	1
16	What will be the output of the following Python code? y = 8 def update(): y = y + 2 print(y) update() a) 10 b) 8 c) Error d) None	1
17	In _____ switching, before a communication starts, a dedicated path is identified between the sender and the receiver. a) Graph b) Plot c) Circuit d) Packet	1
18	What will be the output of the following Python code? text = "Drone Expo" print(text[-2: :-4])	1
19	Which MySQL command correctly removes an attribute Age from table Student? a) ALTER TABLE Student DROP COLUMN Age; b) ALTER TABLE Student REMOVE Age; c) DROP COLUMN Age FROM Student; d) DELETE COLUMN Age FROM Student;	1
Q20 and Q21 are Assertion(A) and Reason(R) based questions. Mark the correct choice as: a) Both A and R are True and R is the correct explanation for A. b) Both A and R are True and R is not the correct explanation for A. c) A is True but R is False. d) A is False but R is True.		
20	Assertion (A): Opening a file in mode rb allows you to read a binary file without modifying it. Reason (R): In rb mode, if the file does not exist, Python creates a new empty file.	1

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21	Assertion (A): The MySQL query SELECT Student_Name FROM Student WHERE Guardian_Phone IS NULL; displays the names of students whose guardian phone number is not recorded. Reason (R): In MySQL, you can also use = to check for NULL values; it is equivalent to IS NULL to identify columns with no value.	1
	Section-B (7 x 2=14 Marks)	
22	A. Define global variable and local variable with a suitable example. OR B. Write the difference between the seek() and tell() methods in Python text file handling with a suitable example.	2
23	The Python program below tries to print the list in reverse. However, there are syntax and logical errors in the code. Rewrite it after removing all the errors. Also, underline all the corrections made words=["Class Test","Term Exams","Pre-board Exam","Board Exam'] L=len(words) for i in range(L, -1, -1) Print (words[i] , end = " ")	2
24	A. Write a Python function capitalize_words(words) that takes a list of strings words and returns a new list with each word capitalize. Example: If the input is ["python", "computer", "science"] The output should be ["Python", "Computer", "Science"]. OR B. Write a Python function sum_positive(list1) that takes a list of numbers and prints the sum of all positive numbers in the list. Example: If the input is [10, -5, 20, -3, 7] The output should be 37.	2
25	A. (Answer using Python built-in methods/functions only): I. Write a statement to check whether the string username starts with "admin". II. Given data = [2, 7, 9, 9, 2], write a statement to count how many times the largest element occurs. OR B. Predict the output of the following Python code: text = "Python Programming" print(text.replace("o", "0")) print(text.find("z"))	2

26	Predict the output of the following Python code:
	code = "gLaD5"
	result = []
	i = 0
	while i < len(code):
	if code[i].isupper():
	result.append(code[i].lower())
		2

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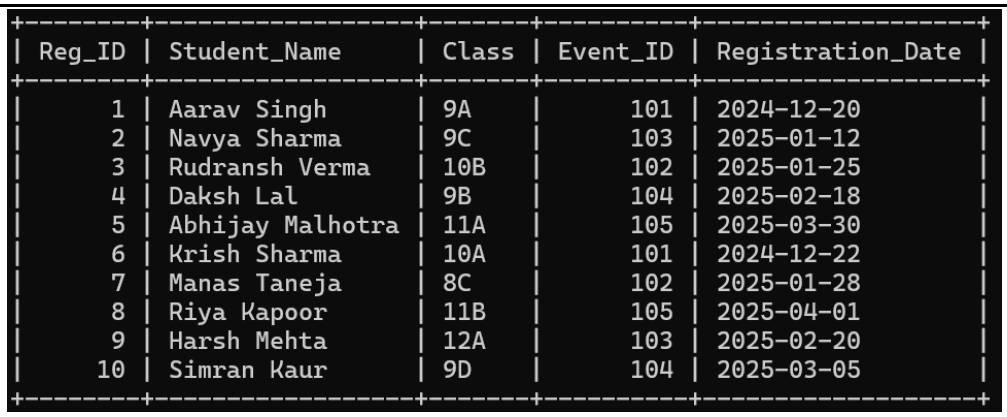
	elif code[i].isdigit():
	result.append(int(code[i]) * 2)
	else:
	result.append("#")
	i += 1
	print(result)	

27	A. Write suitable commands to do the following in MySQL. I. Display a list of all the relations stored in a database. II. Delete a database named CS_SP. OR B. Differentiate between alter and update command in MySQL with a suitable example.	2
28	A. Which protocol helps us to browse through web pages using internet browsers? Name any one internet browser. OR B. Define tree topology. Also, write down any one limitation of tree topology.	2
	Section-C (3 x 3 = 9 Marks)	
29	A. Write and call a Python function to read "History.txt" and display all lines that contain a four-digit year (e.g., 1999, 2024 etc.) OR B. Write a Python function that displays the number of times the word "Python" appears in a text file named "Prog.txt".	3
30	A list containing records of books is given as: B = [("Machine Learning Basics", 550), ("World Geography", 450), ("Wonder World", 140), ("AI Revolution", 750)] Write the following user-defined functions to perform operations on a stack named Books: I. Push_element()**:** Push only those books whose price is more than 250 into the stack. II. Pop_element(): Pop the items from the stack and display them. After all items are popped, display "Stack Empty". Expected Output: ('AI Revolution', 750) ('World Geography', 450) ('Machine Learning Basics', 550) Stack Empty	3
31	A. Write a Python function avg_len( ) segment that adds up the lengths of all the words in a list and then prints the average (mean) length. OR B. Consider the tuple pairs = ((2,5),(4,2),(9,8),(12,10)). Write a Python function cnt_even( ) to count the number of pairs (a, b) where both a and b are even.	3
	Section-D (4 x 4 = 16 Marks)	

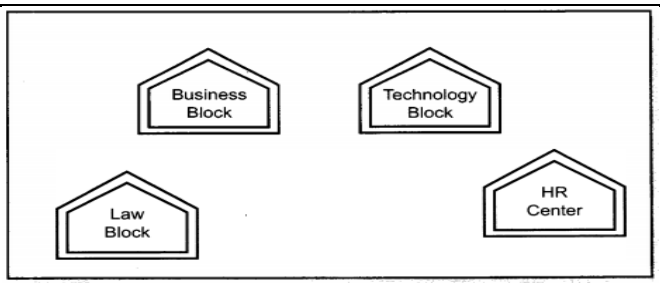
*There is no change in the Question Paper Design and Assessment Pattern for Academic Session 2026-27.

32	Consider the table TECH_EVENT_REG as given below: +--------+------------------+------------------+-------+---------+ | reg_id | participant_name | event_name | grade | reg_fee | +--------+------------------+------------------+-------+---------+ | 501 | Aarav | Hackathon | 11 | 200 | | 502 | Riya | RoboRace | 10 | 150 | | 503 | Krish | WebDev Challenge | 12 | 250 | | 504 | Navya | Hackathon | 11 | 200 | | 505 | Samaira | CodeSprint | 9 | 100 | | 506 | Rohan | RoboRace | 10 | 150 | +--------+------------------+------------------+-------+---------+ A. Write the following queries: I. To display the number of participants in each event where the number of participants is greater than 1. II. To display the distinct event names available in the Tech Event. III. To display details of participants whose names contain the letters 'an'. IV. To display all records of TECH_EVENT_REG sorted by participant_name in descending order. **OR** B. Predict the output of the following: I. Select participant_name, event_name from tech_event_reg where grade = 11; II. Select event_name, count(*) from tech_event_reg group by event_name; III. Select participant_name from tech_event_reg where event_name like '%challenge'; IV. Select max(reg_fee) from tech_event_reg where grade in (9, 10);	4
33	Kabir runs a vehicle rental agency and maintains a CSV file named Rentals.csv. The file stores daily rental details with columns: Vehicle_ID, Vehicle_Type, Hours_Rented, Rate_Per_Hour Create the following user-defined functions: I. AddRental() – to accept rental details from the user and append them to Rentals.csv. II. CalculateTotalRent() – to calculate and return the total rent earned (Hours_Rented × Rate_Per_Hour).	4
34	Aditi is managing an events database and needs to access certain information from the chess_event and registrations tables for an upcoming survey. Help her extract the required information by writing the appropriate MySQL queries as per the tasks mentioned below: Table: Chess_Event +----------+-----------------------------+------------+----------------------+-----------------+ | Event_ID | Event_Name | Event_Date | Venue | Max_Participants | +----------+-----------------------------+------------+----------------------+-----------------+ | 101 | Inter-School Chess Championship | 2025-01-15 | Auditorium Hall | 60 | | 102 | Rapid Chess Tournament | 2025-02-10 | Activity Room 3 | 40 | | 103 | Under-16 Chess Cup | 2025-03-05 | Sports Complex Room 2 | 50 | | 104 | Teachers vs Students Chess Meet | 2025-04-12 | Library Conference Area | 30 | | 105 | Open Blitz Chess Challenge | 2025-05-20 | IT Lab 1 | 80 | +----------+-----------------------------+------------+----------------------+-----------------+ Table: Registrations	4

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	 I. Write a MySQL query to update the venue of the event "Inter-School Chess Championship" to "Multi-Purpose Hall". II. Display the details of all chess events scheduled after 2025-01-01. III. Write a MySQL query to display the registration details of students who have registered for events scheduled in 'Auditorium Hall', 'Activity Room 3', or 'Library Conference Area'. IV. a. Display the student_name along with the event_name for all registrations or b. Write a MySQL query to display the cartesian product of Chess_Event and Registrations tables.	
35	MySQL database named LibraryDB has a table book_records which contains the following attributes: • Book_ID**:** ID of the book (Integer) • Book_Title: Title of the book (String) • Author: Author of the book (String) • Copies: Number of copies available (Integer) Consider the following details to establish Python-MySQL connectivity: • Username: librarian • Password**:** lib@2025 • Host: localhost Write a Python-MySQL program to change the Book_Title to 'Data Structures in Python' for the record whose Book_ID is 109 in the book_records table.	4
	Section-E (2 X 5 = 10 Marks)	
36	A college administrator, Atul, maintains student course enrollment records. Each record includes Roll_No, Student_Name, Course, Fees. Write Python functions to: I. Input enrollment details and append them to a binary file. II. Update the Fees of all students enrolled in the "Data Science" course to 75000.	2+3
37	AKA University is setting up its Academic blocks at Raipur and planning to set up a network. The university has 3 academic blocks and one Human Resource Center as shown in the diagram below:	5

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Center to center distances between various blocks is as follows:

Law Block to Business Block	90 m
Law Block to Technology Block	55 m
Law Block to HR Center	120 m
Business Block to Technology Block	70 m
Business Block to HR Center	105 m
Technology Block to HR Center	45 m

Number of Computers in each of the Blocks/Center is as follows:

Law Block	30
Technology Block	55
HR Center	120
Business Block	45

I. Suggest the most suitable place (i.e. Block/Center) to install the server of this university with a suitable reason.

II. Identify and draw the most suitable topology for connecting all the blocks if the university wants a centralized network management system.

III. Suggest and placement of the following devices with reasons.
   a. Repeater
   b. Switch

IV. The Institute is planning to link its study center situated in Delhi. Suggest a way to connect it with reasonably high speed guided medium.

V. A. The Vice-Chancellor wants to hold a face-to-face digital meeting with the Heads of Departments in all blocks simultaneously. Which Protocol will be used?

OR

B. What is the use of VoIP?

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INFORMATICS PRACTICES (Code No. 065)
SAMPLE QUESTION PAPER (THEORY)
Class - XII (2026-27)

Time Allowed: 3 Hours **Maximum Marks: 70**

General Instructions:

- All questions are compulsory.
- The examination paper contains five sections, from Section A to Section E.
- Section A consists of 21 questions (1 to 21). Each question carries 1 Mark.
- Section B consists of 7 questions (22 to 28). Each question carries 2 Marks.
- Section C consists of 4 questions (29 to 32). Each question carries 3 Marks.
- Section D consists of 2 questions (33 to 34). Each question carries 4 Marks.
- Section E consists of 3 questions (35 to 37). Each question carries 5 Marks.
- There is no overall choice. However, internal choices have been provided in some questions. Attempt only one of the choices in such questions.
- All programming questions are to be answered using Python Language only.
- In case of MCQ, text of the correct answer should also be written.

QNO	Section-A (1 to 21) 21 x 1 = 21 Marks	MARKS
1	A Pandas Series S1 is created with index labels ['a', 'b', 'c', 'd', 'e']. State whether the following statement is True or False: "The statement S1['b':'d'] will produce the same result as S1[1:3]."	**1**
2	Ravi needs to find the total number of characters in the StudentName column. Which SQL function should he use? A. COUNT() B. SIZE() C. LENGTH() D. TOTAL()	**1**
3	Which type of Intellectual Property Right (IPR) protects an invention by giving the creator exclusive rights to make, use, and sell it for a specific period? A. Copyright B. Trademark C. Patent D. Software License	**1**
4	While plotting a graph using Matplotlib, which function is used to specify the label for the Y-axis? A. label_y() B. ylabel() C. yaxis_name() D. set_y()	**1**
5	I am a networking device that converts digital data from a computer into analog signals for transmission, and then converts incoming analog signals back into digital data for the computer to understand. Who am I?	**1**

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	A. Repeater B. Router C. Modem D. Switch	
6	Which of the following is a Data Manipulation Language (DML) command in SQL? A. ALTER B. DROP C. UPDATE D. CREATE	**1**
7	The process of re-selling old electronic goods after minor repairs or modification is called: A. Recycling B. Refurbishing C. Reducing D. Leaching	**1**
8	What will be the output of the following code? import pandas as pd print(pd.Series(5, index=[0, 1, 2])) A. A Series with values 5, 5, 5 B. A Series with value 5 only at index 0 C. A Series with values 0, 1, 2 D. Error	**1**
9	Fill in the blank: The SQL clause used to filter the groups formed by the GROUP BY clause based on a specific condition is _____.	**1**
10	Riya visited an online shopping website, and it saved small text files on her computer to remember her preferences and session details. What are these files called? A. Plug-ins B. Add-ons C. Cookies D. Scripts	**1**
11	Which keyword is used with the ORDER BY clause to sort data in descending order? A. DESC B. DOWN C. DSC D. REVERSE	**1**
12	If we add two Pandas Series that do not have matching index labels, the result for those non-matching indices will be: A. 0 B. Error C. NaN D. Empty string	**1**
13	The process of buying electronic gadgets only when necessary to minimize the generation of electronic waste is an example of which 'R' of e-waste management? A. Recycle	**1**

*There is no change in the Question Paper Design and Assessment Pattern for Academic Session 2026-27.

	B. Reuse C. Reduce D. Refurbish	
14	Which SQL function is used to extract a specific number of characters from a string starting from a given position? A. LEFT() B. RIGHT() C. LENGTH() D. MID()	**1**
15	Aman is exporting a Pandas DataFrame to a CSV file using the to_csv() function. He wants to save only the data and avoid writing the row indices to the file. Which parameter should he set to False to prevent the indices from being included? A. header B. sep C. index D. columns	**1**
16	Fill in the blank: The process of storing a website on a web server so that it can be accessed over the internet is known as _____.	**1**
17	A HR manager queries an employee database. He wants to retrieve only those records where the Commission field actually contains a value. Which SQL condition should he use? A. WHERE Commission != NULL B. WHERE Commission IS NOT NULL C. WHERE Commission <> NULL D. WHERE Commission NOT NULL	**1**
18	When creating a DataFrame from a list of dictionaries, the dictionary keys become: A. Row indices B. Data values C. Column labels D. Positional Index	**1**
19	State whether the following statement is True or False: In SQL, the BETWEEN operator is inclusive of both the lower and upper values specified in the range.	**1**
	Q. 20 and Q. 21 are Assertion (A) and Reason (R) type questions. Choose the correct option as: A. Both Assertion (A) and Reason (R) are True and Reason (R) is the correct explanation for Assertion (A). B. Both Assertion (A) and Reason (R) are True and Reason (R) is not the correct explanation for Assertion (A). C. Assertion (A) is True and Reason (R) is False. D. Assertion (A) is False, but Reason (R) is True.	
20	**Assertion (A)** The statement df.drop('Science', axis=0) will remove the column labeled 'Science' from the DataFrame. **Reason (R)** In Pandas drop() method, axis=0 refers to row-wise operations and axis=1 refers to column-wise operations.	**1**
21	**Assertion (A)** In SQL, the query select * from emp; and SELECT * FROM EMP; will produce the same result. **Reason (R)** SQL is a case-sensitive language.	**1**
	Section-B (22 to 28) 7X2 = 14 Marks	

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22	**(A)** (i) Explain the use of the head() function in a Pandas DataFrame. (ii) What happens if the parameter n is not specified in the head() function? **OR** **(B)** Explain the concept of Boolean Indexing in DataFrames. Write a small Python code example to demonstrate it.	**2**
23	Differentiate between an Active and a Passive digital footprint.	**2**
24	Define Data Visualization and state its primary purpose.	**2**
25	**(A)** Define Star Topology. Mention one major advantage of using it. **OR** **(B)** What is VoIP? State any one advantages of using VoIP for communication.	**2**
26	Write SQL queries for the following requirements: (i) Display the last 3 characters of the attribute Ename from the EMPLOYEE table. (ii) Find the position of the substring 'ma' in the string 'Informatics'.	**2**
27	Mention and briefly explain any two guidelines of netiquette.	**2**
28	**(A)** Consider the following two dictionaries: dict1 = {'a': 10, 'b': 20} dict2 = {'a': 5, 'b': 10, 'c': 20} Write Python code to create a DataFrame from a list containing the above two dictionaries. **OR** **(B)** Mention the purpose of read_csv() and explain the use of its sep parameter.	**2**
	Section-C (29 to 32) 4X3 = 12 Marks	
29	Consider the following digital security scenarios and answer the questions: (i) Mehak receives an SMS from an unknown number stating she has won a lottery. To claim it, she must click a link and enter her Aadhaar number and bank OTP. What type of cybercrime is this? (ii) A group of people repeatedly post insulting comments and spread rumors about a classmate on a social media platform to distress them. What is this activity called? (iii) Suggest one Communication Etiquette that should be followed to maintain credibility while participating in an online discussion forum.	**3**
30	**(A)** (i) Write a Python program to create a Pandas Series named Inventory from the following dictionary: stock = {'Pens': 150, 'Erasers': 80, 'Sharpeners': 60, 'Notebooks': 200}. (ii) Consider two Series S1 and S2: import pandas as pd S1 = pd.Series([10, 20], index=['a', 'b']) S2 = pd.Series([30, 40], index=['b', 'c']) Write the output of the statement: print(S1 + S2) **OR**	**3**
	(B) Consider a DataFrame df as shown below: Name IP Maths	

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	0 Arpit 95 88 1 Vani 89 92 2 Chetan 78 80 Write Pandas statements to perform the following actions: (i) Add a new column 'Total' containing the values [183, 181, 158]. (ii) Rename the 'Name' column to 'Student'. (iii) Delete the 'Maths' column from the DataFrame.	
31	Consider the table HOSPITAL, which contains the following columns: Column Name – Remark PNo – Patient Number PName – Patient Name Dept – Department DOA – Date of Admission Table: HOSPITAL PNo | PName | Dept | DOA 101 | Ravi Kumar | Cardiology | 2024-05-12 102 | Anita Sharma | Neurology | 2025-06-01 103 | Mohan Das | Cardiology | 2023-05-18 104 | Seema Verma | Orthopedics | 2022-04-22 105 | Rahul Singh | Cardiology | 2025-03-30 Write SQL queries to perform the following tasks: (i) Display the names of all patients in the Cardiology department in lowercase. (ii) Display each patient's number along with the day name of the week on which they were admitted. (iii) Display the patient names arranged in alphabetical order.	**3**
32	**(A)** Consider a table named STUDENT containing the following data: ID | Name | DOB 1 | Aman | 2005-08-15 2 | Vinod | 2006-01-26 3 | Saurabh | 2004-01-01 Write the output of the following SQL queries: (i) SELECT UPPER(Name), LENGTH(Name) FROM STUDENT WHERE ID = 2; (ii) SELECT DAY(DOB), YEAR(DOB) FROM STUDENT WHERE ID = 3; (iii) SELECT RIGHT(Name, 3) FROM STUDENT WHERE ID = 1; **OR** **(B)** Write the output of the following SQL queries: (i) SELECT POWER(MOD(13, 5), 2); (ii) SELECT UCASE(MID("Digital India", 9, 5)); (iii) SELECT LENGTH(TRIM(" Informatics "));	**3**
	Section-D (33 to 34) 2X4 = 8 Marks	**MARKS**
33	Write a Python program using Matplotlib to plot a Line Graph showing the "Monthly Sales" of a shop with the following data: Months: ['Jan', 'Feb', 'Mar', 'Apr'] Sales: [12000, 15000, 11000, 18000] The program should include: • Code to import the required library.	**4**

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- Code to plot the line graph with a label "Sales".
- A title "Sales Performance".
- Labels for both X-axis ("Month") and Y-axis ("Amount").
- Code to display the legend and show the plot.

34 **(A)** Consider the following tables: **4**

Table: SPORTS

RegNo	SName	Game	Fees
101	Aryan	Cricket	500
102	Riya	Tennis	600
103	Kabir	Cricket	500
104	Ishaan	Tennis	600
105	Tejas	Football	750

Table: COACH

CoachID	CName	Game
C1	Venkatesh	Cricket
C2	Rameshwar	Tennis
C3	Sridhar	Football

Write SQL queries for the following:
(i) Display the Game name and the total fees collected for each game, but only for those games where the total fees exceed 1000.
(ii) Display the Student Name (SName) and their respective Coach Name (CName).

OR

(B) Consider the following tables:

Table: Employee

EmpID	Name	DeptID	Salary
301	Aarav	D1	5200
302	Riya	D2	7500
303	Kabir	D2	6800
304	Ananya	D3	4000
305	Ishaan	D1	4900

Table: Department

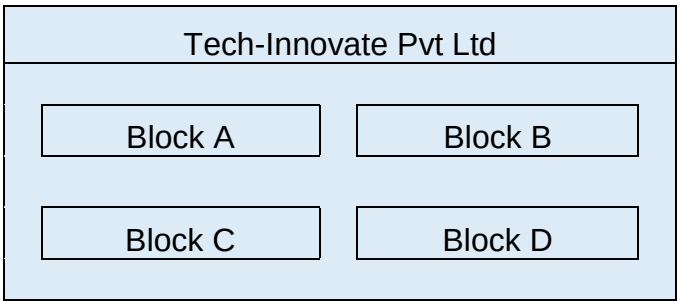
DeptID	DeptName
D1	HR
D2	IT
D3	Sales

Write the output of the following SQL queries:

(i) SELECT DeptID, COUNT(*), MAX(Salary) FROM Employee GROUP BY DeptID HAVING COUNT(*) > 1 ORDER BY DeptID;
(ii) SELECT Name, DeptName FROM Employee, Department WHERE Employee.DeptID = Department.DeptID AND Employee.Salary > 6000;

Section-E (35 to 37) 3X5 = 15 Marks

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35	Tech-Innovate Pvt Ltd is a leading technology solutions provider renowned for delivering innovative digital services to clients worldwide. The company operates from a spacious campus comprising four blocks: Block A, Block B, Block C, and Block D.  To ensure smooth communication, centralized data storage, and high-speed connectivity across the campus, the organization plans to set up a structured network connecting all blocks. The following tables provide the distances between the blocks and the number of computers in each block. From / To / Distance (meters) Block A / Block B / 40 m Block A / Block C / 50 m Block A / Block D / 160 m Block B / Block C / 70 m Block B / Block D / 60 m Block C / Block D / 120 m Block / Number of Computers Block A / 25 Block B / 60 Block C / 15 Block D / 150 As a network engineer, answer the following questions based on the details given above: (i) Suggest the most suitable block to install the server. Justify your answer. (ii) What type of network (PAN, LAN, MAN, or WAN) will be formed within the campus? (iii) Suggest the placement of a Switch with suitable justification. (iv) The company wants to host a website where clients can log in and see real-time project updates. Should they develop a Static or a Dynamic website? Why? (v) Identify between which blocks a repeater is needed and justify your answer.	**5**
36	(i) Create a DataFrame named StoreDF as shown below using dictionary of series: Items / Quantity / Unit_Price 0 Bread 50 30 1 Milk 100 25 2 Eggs 120 6 Write suitable Pandas statements to Perform the following operations: (ii) Add a new row containing data [Butter, 40, 55]	**3+1+1 = 5**

*There is no change in the Question Paper Design and Assessment Pattern for Academic Session 2026-27.

	(iii) Display values of Items column only,	
37	**(A)** Refer to the table given below: Table: BOOK (i) Insert a new record into the table with the values: 205, The White Tiger, Aravind Adiga, 350. (ii) Update the Price of the book with BookID 202 to 325. (iii) Delete the records of all books written by 'R. K. Narayan'. (iv) Add a new column Publisher of type VARCHAR(30) to the table. (v) Drop the table BOOK from the database. **OR** **(B)** Consider a table named PRODUCT, which contains the following columns: PCode, PName, Category, Qty, Price, ExpiryDate. The columns are described as follows: PCode – A unique code assigned to each product. PName – The name of the product. Category – The category or type of the product. Qty – The available quantity of the product. Price – The unit price of the product. ExpiryDate – The expiration date of the product. Write SQL queries for: (i) Display PName and its Price rounded to one decimal place. (ii) Display the names of products that end with the letter 's'. (iii) Display the PName and Category of those products whose Price is less than or equal to 500. (iv) Display the total quantity of all products. (v) Display PName after removing leading spaces only.	**5**

BookID	Title	Author	Price
201	Wings of Fire	A P J Abdul Kalam	350
202	The Guide	R. K. Narayan	299
203	The God of Small Things	Arundhati Roy	399
204	Malgudi Days	R. K. Narayan	280

BIOTECHNOLOGY – CODE NO. 045
SAMPLE QUESTION PAPER*
CLASS – XII (2026-27)

Maximum Marks: 70 **Time allowed: 3 Hrs**

General Instructions:

(i) All questions are compulsory.
(ii) The question paper has five sections.
(iii) Section A contains 16 Multiple choice questions of 1 mark each; Section B has 5 short answer questions of 2 marks each; Section C has 7 short answer questions of 3 marks each; Section D has two case-based question of 4 marks; Section E has three long answer questions of 5 marks each.
(iv) *There is no overall choice. However, internal choices have been provided in some questions. A student has to attempt only one of the alternatives in such questions.*

Q. No.	SECTION A	Marks
1.	An organism has 20% of its genome as genes and the genome is divided into four chromosomes. Identify this organism. A. *Arabidopsis thaliana* B. Humans C. Drosophila D. Yeast	1
2.	Subtilisin is inactivated due to oxidation of amino acid A. histidine B. methionine C. aspartate D. serine	1
3	The charge relay system in serine proteases helps in A. stabilising the substrate by forming disulfide bonds. B. activating the serine residue for nucleophilic attack. C. lowering the pH of the active site. D. transporting electrons through the electron transport chain.	1
4	What is the main characteristic of a turbidostat in a continuous microbal culture system? A. Maintains a constant rate of nutrient feed B. Promotes maximum biomass accumulation C. Increases nutrient supply over time D. Ensures constant cell concentration	1
5	Meristem is used as an explant for : A. Obtaining virus resistant plants B. Promoting somaclonal variations C. Producing commercially important secondary metabolites D. Producing Virus free plants	1

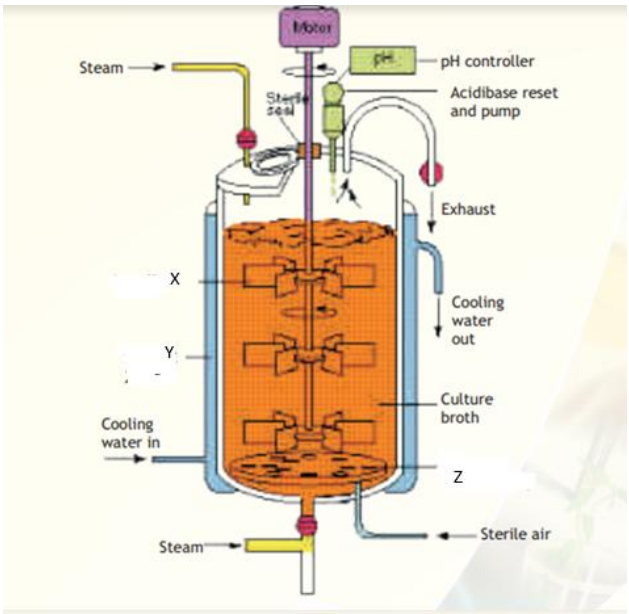
**There is no change in the Question Paper Design and Assessment Pattern for Academic Session 2026-27.*

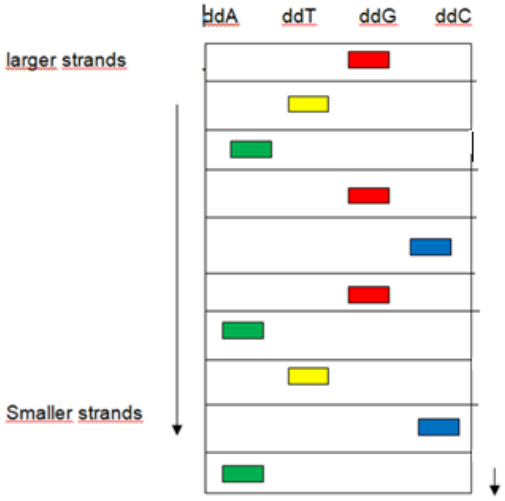
6	A student is interested in amplifying a specific region of genome of an organism. Which of the following would actually help her in targeted amplification? A. Probes B. Primers C. rNTPs D. ddNTPs	1
7	There is no increase in the cell concentration in the lag phase due to the following reason - A. Exhaustion of the medium. B. Space constraint C. Accumulation of toxic products D. Acclimatization to the new environment	1
8	How does the inclusion of phenol red in many cell culture media demonstrate a practical application of understanding the medium's properties? A. Phenol red directly contributes to cell growth by providing essential nutrients. B. Phenol red acts as a visual indicator, simplifying pH monitoring and maintenance of optimal cell culture conditions. C. Phenol red is essential for maintaining the correct osmolality of the medium. D. Phenol red is a necessary cofactor for many enzymatic reactions within the cell.	1
9	Somatic embryos are A. identical with zygotic embryos and without seed coats B. identical with zygotic embryos and with seed coats C. non-identical with zygotic embryos and without seed coats D. non-identical with zygotic embryos and with seed coats	1
10	Which of the following protein pharmaceutical is not produced by CHO cell line? A. Erythropoietin B. Factor IX C. Interleukin 2 D. Monoclonal antibody	1
11	The correct sequence of steps in the technique of Microarray is: (i) Choosing a cell population (ii) Conversion to cDNA molecules	1

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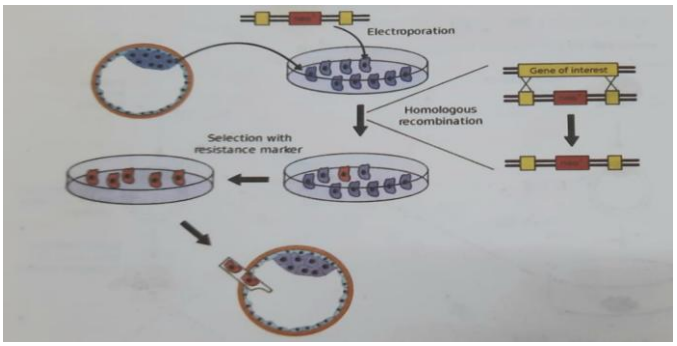
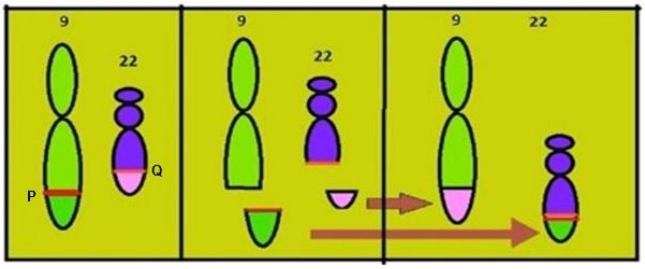
	(iii) Extraction of mRNA molecules (iv) Fluorescent labelling (v) Scanning and interpretation A. (i), (ii), (iv), (iii), (v) B. (i), (ii), (iii), (v), (vi) C. (i), (iii), (iv), (ii), (v) D. (i), (iii), (ii), (iv), (v)	
12	The vectors suitable for transformation of animal cells are: A. SV 40 vectors and papilloma virus vectors B. Cauliflower mosaic virus (CMV) vector and Gemini vectors C. Lambda phage and M13 phage vectors D. T4 phage vectors	1
For Questions number 13 to 16, two statements are given one labelled as **Assertion** (A) and the other labelled as **Reason** (R). Select the correct answer to these questions from the codes (A), (B), (C) and (D) as given below. A. Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). B. Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A). C. Assertion (A) is true, but Reason (R) is false. D. Assertion (A) is false, but Reason (R) is true.		
13	**Assertion**: Proteins have been engineered to show minimum deleterious effects. **Reason**: Proteins are the main molecules which provide stimulus for immunity.	1
14	**Assertion**: The Somatic cell hybridization offers an excellent alternative for obtaining distant hybrids. **Reason**: Somatic hybrids are created by fusing intact cells of distantly related plants.	1
15	**Assertion**: Auxins are added to the culture medium if callus induction is desired. **Reason**: Nature and quantity of auxin added depends on source of explant.	1
16	**Assertion**: Genetic variation between individuals is exploited in DNA fingerprinting. **Reason**: Not all genetic variations are useful in DNA fingerprinting.	1
SECTION B		
17	What is specific growth rate in microbial growth kinetics? Write the relation between specific growth rate and doubling time.	2
18	***Attempt either option A or B.***	2

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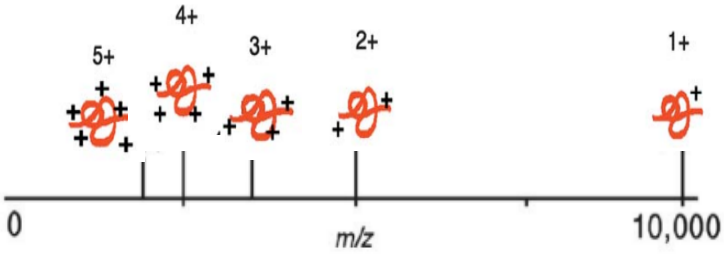
	A. Give any four requirements to carry out the polymerase chain reaction. **OR** B. Describe any two methods to directly introduce foreign DNA into the host cells.	
19	A researcher is using baffle flasks for small scale microbial culture in the lab. How are baffle flasks different from ordinary flasks?	2
20	What is protoplast culture? Write any two of its uses.	2
21	Explain the key difference between polyclonal and monoclonal antibodies in terms of their binding specificity to antigens. How does this difference arise from their cellular origins?	2
	SECTION C	
22	A kidney transplant patient is prescribed OKT3 to prevent acute rejection. Explain how OKT3 modifies T-cell activity?	3
23	Write about any three ways that allow us to perform gene analysis using Bioinformatics.	3
24	A. Identify the parts of fermenter which are labelled as X, Y and Z.  B. "A transgene may be added, which encodes an enzyme to modify a metabolite produced by the organism to yield a new product of interest". Will this be considered as strain improvement? Justify giving an example. - - - - - - - - - - - - - - - - - - - *For Visually impaired students* A. What is the capacity of a small scale biofermentor used in laboratories? Mention any two uses of such bioreactors. B. "A transgene may be added, which encodes an enzyme to modify a	3

	metabolite produced by the organism to yield a new product of interest". Will this be considered as strain improvement? Justify giving an example.	
25	***Attempt either option A or B.*** A. A group of Indian scientists observed that group of patients feeding on whey are responding better to retro-viral therapy as compared to those who are not taking it. What could be the possible scientific explanation for this therapeutic effect? **OR** B. Write about the three parameters that define the dietary value of proteins.	3
26	A. Illustrate the steps involved in the generation of RFLP. B. What is the RFLP pattern expected in case of identical twins.	3
27	 Look at the diagram and answer the following questions A. Derive at the original DNA sequence by reading above given autoradiogram. B. Why are ddNTPs used along with dNTPs in the enzymatic method of DNA sequencing? C. How is a dNTP chemically different from a ddNTP? - - - - - - - - - - - - - - - - - - - *For Visually impaired students* The sequence of the strand of DNA complementary to the sequenced strand is 5' ACGCCCGAGTAGCCCAGA 3' A. Write the sequence of the original strand. B. Which method is commonly used to sequence DNA? C. Why is automated DNA sequencing considered safer?	3

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28	 A. Based on the visual cues, Identify the technique which is depicted here? B. Expand ICM. From which developmental stage is it obtained, and why is this stage ideal for pluripotency research? - - - - - - - - - - - - - - - - - - - *For Visually impaired students* A. Differentiate between finite and continuous cell lines based on their lifespan in culture. B. Herceptin is a monoclonal antibody used to treat HER2 - positive breast cancer. Based on the information provided, what is the role of the HER2 receptor in cancer cell growth, and how does Herceptin interfere with this process to inhibit cancer progression?	3
	SECTION D	
29	Chronic Myelogenous Leukemia is caused due to Reciprocal translocation mutation between chromosome 9 and 22 that forms an extra-long chromosome 9 and extra small chromosome 22 containing the fused gene regions. Refer to the schematic view representing metaphase chromosome and answer the question that follow:  A. Identify the gene region P and Q in the given diagram. B. What is the extra small resultant chromosome 22 called? *Attempt either subpart C or D.* C. Apart from Karyotyping, which other technique can be used to detect CML in patients? State the principle of the technique. **OR** D. Why is Karyotyping not, a preferred method to find out the severity of CML disease? - - - - - - - - - - - - - - - - - - -	4

**There is no change in the Question Paper Design and Assessment Pattern for Academic Session 2026-27.*

	For Visually impaired students Chronic Myelogenius Leukemia is caused due to reciprocal translocation between chromosome 9 and 22 that forms an extra-long chromosome 9 and an extra small chromosome 22 containing the fused gene regions. Both the chromosomes are metaphase arrested and visible during Karyotyping. Answer the questions that follow: A. What are the names given to both the gene regions of chromosome 9 and 22 that fuse? B. What is the extra small chromosome 22 known as? *Attempt either subpart C or D.* C. Apart from Karyotyping, which other technique can be used to detect CML in patients? What is its principle? **OR** D. Why is Karyotyping not, a preferred method to find out the severity of CML disease?	
30	Mass spectrometry  (MS) has emerged as an important tool for characterisation. A protein with a molecular weight of 10,000 dalton generates five different peaks with the ions containing 5, 4, 3, 2, and 1 charges, respectively, as shown below. A. What happens if there is a loss of charge from a biomolecule? B. Mass spectrometry is an analytical tool. Justify the statement. *Attempt either subpart C or D.* C. Calculate the m/z ratio each for protein ions containing 4 and 2 charges. OR D. A protein has a molecular weight of 20,000 daltons and it forms two protein ions containing 6 and 7 charges, what will be its mass/charge ratio? - - - - - - - - - - - - - - - - - - - *For Visually impaired students* Mass spectrometry (MS) has emerged as an important tool in biotechnology. It is extremely useful in obtaining protein structural information such as peptide mass or amino acid sequences. The	4

	molecular ions are generated either by a loss or gain of a charge (e.g. electron ejection, protonation or deprotonation). After the ions are formed, they can be separated according to their m/z ratio and finally detected. A. What happens if there is a loss of charge from a biomolecule? Give two reasons. B. Mass spectrometry is an analytical tool. Justify the statement. *Attempt either subpart C or D.* C. Name the major components of a mass spectrometer. **OR** D. How is MALDI used to volatilize and protonate proteins?	
SECTION E		
31	***Attempt either option A or B.*** A. Few restriction enzymes break the phosphodiester bond in such a manner that single stranded overhang ends are generated in the DNA strand. *EcoRI* is one such a restriction enzyme. I. Write the sequence for restriction site for enzyme EcoRI. Give a name to the kind of ends generated here. II. If the following DNA strand is digested by EcoRI, how many DNA fragments will be generated. Write the sequence of these double stranded DNA fragments with their respective polarity. 5' GTATTAGCTGAATTCTATAAAACG 3' 3' CATAATCGACTTAAGATATTTTGC 5' III. Why is *EcoRI* named so? **OR** B. A gene X was introduced into cloning vector pUC19 at LacZ gene site. What will be the impact on the recombinant plasmids? Also suggest a possible way by which we can differentiate recombinants from non-recombinant plasmids.	5
32	***Attempt either option A or B.*** A. I. A company plans to release a new insect-resistant GM crop. Evaluate three potential bioethical concerns in plant genetic engineering. II. What was the need to develop "Flavr Savr"? **OR** B. I. How can the transgenic plants be used as factories to produce biodegradable plastics? II. What are artificial seeds and mention any one application? Explain diagrammatically.	5

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33	***Attempt either option A or B.*** A. I. Describe the technique developed by O'farrell to study proteomics. II. Name any two properties that can be manipulated using protein Engineering. **OR** B. I. Why is sickle cell anaemia called a molecular disease? II. Who developed peptide mapping? Describe the technique.	5

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**There is no change in the Question Paper Design and Assessment Pattern for Academic Session 2026-27.*